# Volunteering matching application for a crisis management platform

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# Abstract

When disaster strikes, relief organisations face a surge of spontaneous volunteers whose skills and availability are unknown, and turning that goodwill into effective help falls to coordinators who must match hundreds of unfamiliar people to rapidly changing tasks under time pressure and incomplete information. Existing approaches tend toward one of two poles, manual tools that keep coordinators in full control but place a heavy load on their attention, or automated optimisation models that allocate quickly but leave less room for the human oversight crisis work often calls for, with little in between that can adapt to a task's actual risk. This thesis addresses the research question: how can we design a volunteer–task matching application for improved crisis response? Following a Design Science Research methodology and grounded in Cognitive Load Theory, it presents CrisisMatch, a web-based prototype built around a three-tier Human-in-the-Loop architecture that makes the degree of automation a per-task choice, letting coordinators move between manual, semi-automatic, and automatic modes according to task risk. It was evaluated through twelve semi-structured interviews, eight with coordinators and four with volunteers, supported by NASA-TLX ratings. Coordinators found this consolidated approach generally less demanding than their current tools. The findings also reframe what such a system offers. Its value lies less in speed than in adapting to organisations with very different risk profiles. Coordinator trust rested less on the ranking than on the prospect of shaping its criteria, and volunteer coordination proved as much a relational task, where goodwill must be preserved, as a cognitive one.

## Zusammenfassung

Wenn eine Katastrophe eintritt, stehen Hilfsorganisationen vor einer Welle spontaner Freiwilliger mit unbekannten Fähigkeiten und Verfügbarkeiten. Diese Hilfsbereitschaft in effektive Unterstützung zu übersetzen, liegt bei den Koordinierenden, die unter massivem Zeitdruck und mit unvollständigen Informationen hunderte fremde Personen schnell wechselnden Aufgaben zuweisen müssen. Bestehende Ansätze bewegen sich meist zwischen zwei Extremen: manuelle Herangehensweisen, die volle Kontrolle zulassen, aber die Aufmerksamkeit stark beanspruchen, und automatisierte Optimierungsmodelle, die zwar schnell zuteilen, aber kaum Raum für die in der Krisenarbeit wichtige menschliche Aufsicht lassen. Dazwischen gibt es kaum Lösungen, die sich an das tatsächliche Risiko einer Aufgabe anpassen. Diese Arbeit geht der Forschungsfrage nach, wie sich eine Anwendung zur Koordination von Freiwilligen und Freiwilligenarbeit gestalten lässt, um die Krisenreaktion zu verbessern. Folgend dem Ansatz der Design Science Research (DSR) und fundiert in der Theorie der kognitiven Belastung präsentiert diese Arbeit CrisisMatch, einen webbasierten Prototypen auf Basis einer dreistufigen Human-in-the-Loop-Architektur, die den Grad der Automatisierung für jede Aufgabe einzeln wählbar macht. Koordinierende können so je nach Risikostufe zwischen den drei Modi wechseln. Die Evaluierung erfolgte durch zwölf teilstrukturierte Interviews, acht mit Koordinierenden und vier mit Freiwilligen, unterstützt durch NASA-TLX-Bewertungen. Die Koordinierenden empfanden diesen Ansatz als generell weniger belastend als die bisherige Praxis. Die Ergebnisse rücken den Nutzen in ein neues Licht: Der Wert liegt weniger in der reinen Geschwindigkeit als in der Anpassungsfähigkeit an unterschiedliche Risikoprofile. Das Vertrauen basierte weniger auf dem Ranking selbst als auf der Aussicht, dessen Kriterien zukünftig aktiv mitzugestalten. Zudem zeigte sich, dass Freiwilligenkoordination neben der kognitiven Leistung eine ebenso starke soziale Komponente besitzt, da die Hilfsbereitschaft bewahrt werden muss.

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# List of Acronyms

| | |
|---|---|
| **AI** | Artificial Intelligence |
| **API** | Application Programming Interface |
| **CLT** | Cognitive Load Theory |
| **DFA** | Dynamic Function Allocation |
| **DOA** | Degree of Automation |
| **DSR** | Design Science Research |
| **HCI** | Human-Computer Interaction |
| **HITL** | Human-in-the-Loop |
| **JWT** | JSON Web Token |
| **NASA-TLX** | NASA Task Load Index |
| **ORM** | Object-Relational Mapping |
| **SA** | Situation Awareness |
| **SVCP** | Spontaneous Volunteer Coordination Problem |
| **UI** | User Interface |
| **UX** | User Experience |
| **UZH** | University of Zurich |
| **VMS** | Volunteer Management System |
| **VRC** | Volunteer Reception Center |

# 1 Introduction

## 1.1 Motivation & Context

Crisis situations, ranging from natural disasters like floods or fires to rapidly evolving geopolitical events, frequently trigger immediate and massive waves of solidarity within civil society (Whittaker et al., 2015). In these acute moments, official relief organisations and municipalities are typically supported by a rapid influx of spontaneous volunteers. These are individuals who are not formally affiliated with any recognised civil protection agency prior to the event, but who are eager to offer their physical labour or specialised knowledge (Betke et al., 2023; Reuter et al., 2013). This group can also include individuals who have participated in previous crisis response efforts, yet remain outside of official, permanent volunteer registries (Whittaker et al., 2015). Historically, this surge in civic engagement has proven to be an immensely valuable resource. Spontaneous volunteers are often the very first to arrive at a crisis scene, providing critical assistance while official responders are still mobilising or navigating disrupted infrastructure (Betke et al., 2023). However, integrating this massive, unstructured workforce into official disaster management operations presents a severe logistical challenge. Coordination becomes exceptionally difficult at scale due to the inherent unpredictability of the situation. Spontaneous volunteers appear suddenly, their availability fluctuates constantly, and the operational tasks they are needed for can change rapidly under time pressure (Reuter et al., 2013; Sperling & Schryen, 2022). Unlike trained personnel within official aid organisations, spontaneous volunteers do not arrive with pre-verified skills, formal training, or pre-assigned roles. Official organisations are consequently forced to assess the diverse capabilities of hundreds or even thousands of unknown individuals on the spot. This creates a high-pressure environment where disaster managers must safely and efficiently deploy willing citizens without fully knowing their backgrounds or limitations (Sperling & Schryen, 2022; Zettl et al., 2017).

While the volume of volunteers represents great potential for managing disaster impacts, the lack of structured intake and assessment mechanisms makes it incredibly difficult for organisations to absorb this help effectively at scale (Zettl et al., 2017).

## 1.2 Problem statement

A core bottleneck lies in the matching process itself. Coordinators must allocate volunteers to tasks while considering constraints such as skills, time, and availability, which becomes difficult to do consistently when information is incomplete and conditions change (Sperling & Schryen, 2022; Falasca et al., 2009).

Current manual coordination practices, often relying on fragmented communication tools like ad hoc messaging groups, paper lists, or basic spreadsheets, rapidly overwhelm the cognitive capacity of crisis managers. As one interviewee noted during the requirement interview analysis for this thesis, managing communication across multiple platforms quickly becomes unmanageable: "It was just too messy. It was like a lot of these channels, a lot of notifications, and it was not possible to keep up with all of them" (V_Helper_I02: 0:09:39). This reliance on manual, fragmented information processing leads to critical failures. Firstly, it causes significant delays in deployment, leaving willing individuals idle while critical tasks remain unstaffed, and secondly, it results in suboptimal assignments and duplicate bookings (Betke, 2018). This lack of centralised task status tracking causes severe frustration for volunteers, with one interviewee recalling: "I made the half an hour trip for my own money... to only realise that there was a translator after all... for people who provide the help, it's important to know that their help is needed. Otherwise, it can be very frustrating" (V_Helper_I02: 0:34:00).

To solve this, research highlights the need for decision support technologies that can assist in assigning volunteers to tasks while balancing competing objectives (Falasca et al., 2009; Sperling & Schryen, 2022). However, crisis coordination practices differ across contexts, and full algorithmic automation is often viewed critically by practitioners who require situational awareness and human oversight, especially for sensitive tasks (Betke et al., 2023).

Consequently, coordinators often find themselves between two ends of a spectrum: largely manual systems that can cause cognitive overload, and fully automated theoretical models that tend to overlook the trust and contextual nuances required in real-world crisis management.

## 1.3 Research Question

The challenges outlined above lead to the following research question:

> **RQ1:** How can we design a volunteer–task matching application for improved crisis response?

## 1.4 Goal & contributions

To answer this, the thesis proposes CrisisMatch, a web-based prototype that implements a 3-tier Human-in-the-Loop architecture, allowing coordinators to switch dynamically between manual, semi-automatic, and fully automatic assignment modes based on situational demands and task risk. The system consolidates registration, task management, and assignment into a single coordinator dashboard, grounded in Cognitive Load Theory and HCI design principles to reduce the extraneous load imposed by fragmented tools, directly bridging the gap between systems that overwhelm coordinators and those that lack the contextual trust real-world crisis response demands.

# 2 Background

## 2.1 Spontaneous Volunteers in Crisis Response

Before evaluating the existing landscape of volunteer coordination tools, it is necessary to establish the core terminology that defines this domain. Disaster response relies on a complex ecosystem of actors, and the categorisation of these actors dictates how they are managed during a crisis. Within emergency management literature, volunteers are traditionally divided into two broad categories based on their relationship with formal relief organisations: affiliated and spontaneous volunteers.

Affiliated volunteers are individuals attached to recognised voluntary agencies (e.g., Médecins Sans Frontières, Samariterbund) who have received prior training for disaster response and operate within pre-established management structures (FEMA, 2014). Because their capabilities and availability are known prior to an event, their deployment is highly structured.

Spontaneous volunteers (sometimes referred to as unaffiliated or convergent volunteers) are individuals who offer assistance in the immediate aftermath of a disaster without prior affiliation to recognised disaster relief organisations (FEMA, 2014; Zettl et al., 2017). They possess a wide range of unknown skills and arrive ad-hoc, making their capabilities and mass movements highly unpredictable. Furthermore, spontaneous volunteers are sub-divided into two distinct types based on how they provide assistance: digital and physical volunteers. Digital volunteers utilise information systems remotely to aggregate, analyse, and distribute crisis information, often operating far away from the physical disaster zone. Spontaneous unaffiliated on-site volunteers, on the other hand, are physical volunteers who converge at local response sites to offer direct labour, engaging in tangible relief activities such as filling sandbags, providing care, or translating on site (Betke, 2018). While the matching logic of the volunteer coordination system can theoretically be extended to route remote digital tasks, the scope of this thesis is limited to the coordination of physical volunteers.

## 2.2 Matching Approaches

### 2.2.1 Manual & Ad-Hoc

Historically, emergency management has relied heavily on traditional manual frameworks to harness this civic engagement. According to the Federal Emergency Management Agency (FEMA, 2014), the standard approach involves establishing physical Volunteer Reception Centers. At these physical sites, coordinators rely on paper forms to register arriving individuals and use manual coordination processes to assign them to specific disaster relief tasks. While this physical infrastructure provides a centralised gathering point, practical experiences and empirical interview data reveal that these traditional solutions are severely slow and cumbersome in real-world scenarios.

To mitigate the slowness of paper-based systems, organisations frequently transition to ad-hoc digital tools such as spreadsheets. However, relying on these basic digital solutions introduces significant logistical flaws. These methods result in problems described in the literature as overwhelmed coordinators (Betke, 2018) and the absence of a central coordination platform (Reuter et al., 2013), which in practice manifest as slow data intake, fragmented storage, and a lack of real-time situational awareness. The manual matching of unverified volunteer skills to rapidly changing crisis needs creates a severe operational bottleneck. This manual process places an impossibly high cognitive load on decision-makers who must mentally juggle hundreds of variables at once.

Because human cognitive capacity is inherently limited, this bottleneck frequently leads to suboptimal assignments. For example, highly skilled medical personnel might be mistakenly assigned to basic logistics tasks, while other eager volunteers are left standing idle because coordinators cannot process their information fast enough (Falasca et al., 2009; Betke, 2018). Rauchecker & Schryen (2018) provide empirical evidence demonstrating that complex coordination and matching decisions can hardly be handled manually by human responders in comparable quality, especially under the severe time pressure of a disaster. Compounding the failures of these traditional systems is the reality that ad-hoc volunteering has fundamentally evolved in the digital age. Citizens no longer wait for physical reception centres to open. Instead, they self-organise digitally on social media platforms such as Twitter during crises. Starbird & Palen (2011) describe these digital participants as "voluntweeters," highlighting how citizens create a massive informal digital response network (Starbird & Palen, 2011; Reuter et al., 2013). However, while this rapid decentralised self-organisation generates a high volume of activity, it severely lacks structural integration with formal emergency management. This informal digital response often produces overwhelming amounts of unstructured data, creating significant challenges regarding information verification, reliability, and trust. Because these self-organised digital networks operate outside of official command structures, they fail to provide a systematic and safe way to deploy volunteers to physical disaster sites.

Ultimately, neither traditional paper-based frameworks, ad-hoc digital tools such as spreadsheets, nor informal social media networks can adequately meet the complex logistical demands of modern disaster response.

### 2.2.2 Algorithmic & Optimisation

To overcome the severe limitations of manual and ad-hoc coordination, computer scientists and operations researchers have attempted to solve the volunteer matching problem purely through advanced mathematics. These academic approaches treat disaster response as a complex resource allocation challenge that can be optimised using sophisticated algorithms. By translating human capabilities and crisis demands into mathematical variables, researchers aim to automate the assignment process and eliminate the cognitive bottleneck experienced by human coordinators.

One of the foundational mathematical approaches to this problem is the bi-criteria integer programming model developed by Falasca et al. (2009). This classic optimisation model is designed to balance two competing operational objectives simultaneously. On one hand, the algorithm seeks to minimise task shortages by ensuring that critical relief activities receive the necessary manpower. On the other hand, it actively attempts to maximise volunteer preference satisfaction by assigning individuals to roles and time slots that align with their specific skills and availability (Falasca et al., 2009). Building upon this foundation, researchers have developed even more intricate mixed-integer linear optimisation frameworks to handle the dynamic nature of disaster environments.

Rauchecker & Schryen (2018) introduced a comprehensive decision support model that explicitly addresses the coordination of spontaneous volunteers through complex scheduling algorithms. Their work is integrated into the KUBAS system, which is a specialised volunteer coordination platform designed to process continuous influxes of new volunteers and shifting task priorities in real time. Sperling & Schryen (2022) further advanced this mathematical approach by formulating the Spontaneous Volunteer Coordination Problem (SVCP). The SVCP uses lexicographically ordered objective functions to navigate resource scarcity, effectively calculating the mathematically optimal distribution of volunteers across multiple priority classes.

Beyond linear programming, scholars have also explored the concept of dual decision-making in volatile environments. Chen et al. (2021) highlight the necessity of "two-sided matching" under uncertain information. Their model recognises that disaster scenarios lack perfect data, meaning that choices must be informed by both objective operational metrics and subjective human preferences. By structuring the allocation as a mutual matching process, their algorithm attempts to satisfy both the distinct demands of the rescue agencies and the complex motivations of the volunteer teams simultaneously.

Other advanced computational methods leverage network structures to optimise deployments. Kumar & Zaveri (2017) utilise graph-based resource allocation techniques, specifically employing a maximum bipartite graph approach, to efficiently distribute emergency resources during disaster times. This graphical method maximises the assignment of available resources to given tasks while prioritising fairness and execution speed. Windsperger (2025) advocates for a graph-based approach to volunteer management to enhance matching, network exploration, and goal-driven engagement. By mapping entities like skills, tasks, and social relationships into a highly interconnected graph database, this system moves beyond rigid task assignments to enable intelligent, semantic-based matching that fosters long-term volunteer commitment.

Despite the impressive mathematical rigour of these optimisation models, their practical application during actual emergencies remains problematic. These algorithms require precise, structured, and continuously updated input data to function correctly, whereas disaster scenarios are characterised by missing and uncertain information (Chen et al., 2021). Beyond data quality, these mixed-integer linear models can become slow when scaled to thousands of spontaneous volunteers, especially for less critical objectives, where the solver often runs out of time before finding a good solution (Sperling & Schryen, 2022).

### 2.2.3 VMS & Prototypes

To operationalise volunteer coordination, various software platforms have been developed, broadly split into commercial Volunteer Management Systems (VMS) and dedicated crisis prototypes. Although commercial systems are widely used for the day-to-day operations of nonprofits, they expose critical shortcomings in high-stress disaster scenarios. In their survey of solutions such as Volgistics and BetterImpact, Schönböck et al. (2016) identify three core limitations. First, limited profile coverage means these systems fail to capture the dynamic skill sets disasters demand. Second, they operate as isolated proprietary "data islands" that cannot share volunteer information across relief organisations (Schönböck et al., 2016; Kapsammer et al., 2021). Finally, they provide suboptimal allocation effectiveness.

Recognising these gaps, researchers have developed dedicated crisis prototypes that integrate spontaneous volunteers through mobile technology. Auferbauer et al. (2016), for instance, built "CrowdTasker," which broadcasts small, geo-located tasks directly to volunteers' smartphones, while Zettl et al. (2017) focused on communication between authorities and citizen groups. Yet these prototypes sit at the opposite extreme. They are agile and fast to mobilise, but rely on skill- and location-based broadcasting rather than the advanced, multi-objective matching of academic models.

## 2.3 Gap & implications

To bridge this gap, disaster coordination systems must transcend the binary manual versus automatic paradigm. The primary implication derived from this literature review is the necessity of a flexible Human-in-the-Loop (HITL) architecture. In highly volatile disaster environments, purely automated systems often fail because they lack the qualitative, contextual awareness required to make nuanced safety decisions. This makes human coordinators less likely to trust them, especially when those coordinators are already working with highly uncertain and incomplete information (Chen et al., 2021). Yet leaning fully on humans is no better, since human cognition on its own is quickly overwhelmed by the high volume of data typical of a crisis response (Betke, 2018; Reuter et al., 2013).

A flexible HITL approach resolves this paradox by dividing the labour. It offloads the computational heavy lifting of data filtering to the system while giving the human coordinator the power to dynamically delegate authority. Rather than being forced into a single rigid paradigm, the coordinator remains in the loop by determining the appropriate level of system autonomy. They can retain strict, manual approval for high-risk or complex tasks while safely deploying fully automated routing for low-risk, commodity tasks until a predefined shift capacity is reached. By operationalising a multi-tier automation workflow that allows coordinators to dynamically slide between manual control, semi-automatic task proposals, and fully automatic assignment based on real-time task risk, this research proposes a testable prototype designed to explicitly reduce cognitive load without sacrificing human oversight.

## 2.4 HCI foundations

### 2.4.1 Cognitive Load Theory

A foundational explanation for why crisis coordinators become overwhelmed lies in Cognitive Load Theory (CLT), originally developed by Sweller (1988) and elaborated in subsequent work by Sweller (2011).

The theory grounds itself in a well-established model of human cognitive architecture: a working memory that is severely limited in both capacity and duration when processing novel information, and a long-term memory that can store vast amounts of organised knowledge in the form of schemas. Because working memory can only process a very small number of novel elements simultaneously, any task requiring the concurrent manipulation of many interacting information elements risks exhausting this limited resource entirely (Sweller, 2011).

Central to CLT is the concept of element interactivity, the degree to which the elements of a task must be processed simultaneously rather than independently. When element interactivity is low, individual pieces of information can be handled one at a time. When it is high, however, no single element can be understood in isolation, and all must be held in working memory at once, directly multiplying cognitive load (Sweller, 2011). Crisis coordination is a textbook example: a coordinator cannot assign one volunteer without simultaneously considering that volunteer's skills, availability, location, the task's requirements, current shift capacity, and the status of all other pending assignments. Each element interacts with every other, making the task intrinsically demanding regardless of how it is presented.

Sweller (2011) distinguishes between two categories of cognitive load that are directly relevant to interface design. Intrinsic load arises from the inherent complexity of the subject matter, determined by element interactivity relative to the expertise of the person handling the task, and cannot be reduced by design alone. Extraneous load, in contrast, is imposed by the way information is presented rather than by the task itself, and actively competes with intrinsic load for the same limited working memory resources.

This means extraneous load is unnecessary for decision-making and can be reduced through better design, making it the primary lever available to interface designers working in cognitively demanding domains. This insight has been explicitly extended into the field of Human-Computer Interaction. Hollender et al. (2010) demonstrate that established HCI usability principles, such as avoiding split-attention, eliminating redundancy, and keeping displays simple, are in fact direct operationalisations of the

CLT goal of reducing extraneous load at the software level. Their work establishes that the design of the software interface itself constitutes a significant, controllable source of cognitive load, and that reducing it frees working memory for the actual decision task at hand. Mazza (2017) extends this argument to the visual design layer, showing that structured visual hierarchies, contextual cues of place, space, and activity, and the chunking of information actively support an interactor's memory and reduce the cognitive effort required to maintain an accurate picture of operational state.

A closely related consequence of high extraneous load is degraded situational awareness, which Endsley (1995) defines as the perception of relevant elements in the environment, the comprehension of their meaning, and the projection of their status in the near future. When coordinators are forced to reconstruct operational status from fragmented sources, working memory is consumed before a coherent picture can form. Good interface design therefore serves a dual purpose: it reduces unnecessary cognitive overhead while simultaneously supporting situational awareness by making the relevant state of the operation easier to perceive and integrate (Endsley, 1995; Mazza, 2017).

This distinction has direct implications for volunteer coordination systems. The intrinsic load of managing dozens of simultaneous assignments under time pressure is already high and unavoidable. The tools coordinators currently rely on, such as fragmented messaging channels, spreadsheets, and paper lists, add substantial extraneous load on top of it, since the operational picture must be reassembled by hand rather than read off a single source. The design response proposed in this thesis directly targets this extraneous burden: by consolidating operational state into a single glanceable dashboard, automating shift capacity tracking, and offering multiple modes for handling candidate matches, the system offloads the information-processing steps that generate extraneous load, leaving coordinator working memory available for the genuinely complex, high-stakes judgements that require human oversight.

### 2.4.2 Automation Design

Automation is not a binary condition but rather a continuum spanning from fully manual to fully automatic operation. Parasuraman et al. (2000) proposed a foundational model, see Figure 1, in which automation can be applied across four functional stages of human information processing: (1) information acquisition, (2) information analysis, (3) decision and action selection, and (4) action implementation. Within each stage, the degree of automation (DOA) ranges across multiple levels, from systems that merely present data to the operator up to systems that act entirely autonomously without human input.

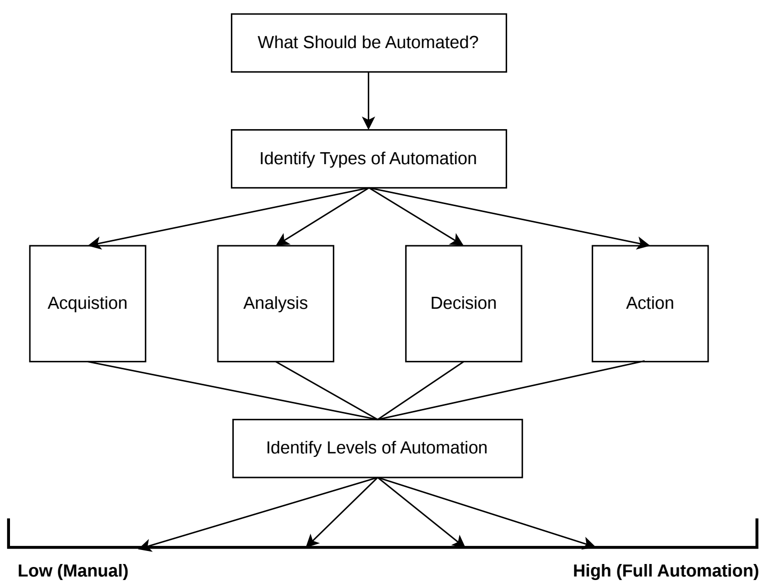


Figure 1: Types and levels of automation. Adapted from Parasuraman et al. (2000)

Different stages of a single system can operate at different automation levels simultaneously, meaning automation design is a multidimensional problem rather than a single dial to turn up or down (Parasuraman et al., 2000). The selection of an appropriate DOA involves a fundamental cost-benefit trade-off. Onnasch et al. (2014) conducted an integrated meta-analysis of 18 studies and confirmed what they termed the "lumberjack effect": increasing the DOA improves routine system performance and reduces operator workload, but simultaneously degrades performance when automation fails and erodes situation awareness (SA).

Their analysis further identified a critical boundary between Stage 2 (information analysis) and Stage 3 (action selection): when automation crosses this threshold by removing the human from the decision-making loop, the negative consequences become substantially amplified. Specifically, return-to-manual performance dropped significantly and loss of SA increased sharply in studies that varied DOA across this boundary. A key mechanism underlying these costs is the out-of-the-loop unfamiliarity problem. When automation consistently handles decision and action selection, operators gradually lose both situational awareness and the manual skills required to intervene effectively. Parasuraman et al. (2000) documented three compounding consequences of high-level automation: reduced situation awareness, complacency (over-trust leading to reduced monitoring), and skill degradation through disuse. Together, these effects mean that the operator who is most needed in a moment of system failure is precisely the operator who is least prepared to act. This creates a paradox in automation design: the higher the DOA, the better the system performs under normal conditions, yet the more fragile the overall human-machine system becomes when conditions deviate from normal (Onnasch et al., 2014; Parasuraman et al., 2000).

Dynamic Function Allocation (DFA) has emerged as a design response to this paradox. Rather than fixing the automation level at design time, DFA adjusts the allocation of functions between human and machine in response to changing situational demands. Atashfeshan et al. (2021) proposed a novel DFA method centered on two questions: when should reallocation occur (the trigger mechanism), and how should it occur (the allocation strategy). Using Bayesian networks to predict human-machine system reliability under uncertainty, their method dynamically reallocates functions based on predicted performance degradation, keeping human operators involved in cognitively critical tasks such as diagnosis and decision-making even during high automation periods. In a controlled experiment simulating a gas power plant control room, the proposed DFA approach improved system reliability by approximately 30% and increased operator SA by approximately 40% compared to static automation, with the strongest effects observed under time pressure and adverse environmental conditions. Taken together, these findings converge on a consistent design principle: automation should be applied at high levels to information acquisition and analysis where benefits are clear and failure consequences are manageable, while human involvement must be preserved at the decision and action selection stages, particularly in safety-critical and unpredictable environments. This principle is directly operationalised in the prototype developed in this thesis through a three-tier automation workflow with each mode mapped onto the levels of automation of decision and action selection shown in Table 1.

| | **Level** | **Description** |
|---|---|---|
| High | 10 | The computer decides everything, acts autonomously, ignoring the human |
| | 9 | informs the human only if it, the computer, decides to |
| | 8 | informs the human only if asked, or |
| | 7 | executes automatically, then necessarily informs the human, and |
| | 6 | allows the human a restricted time to veto before automatic execution, or |
| | 5 | executes that suggestion if the human approves, or |
| | 4 | suggests one alternative |
| | 3 | narrows the selection down to a few, or |
| | 2 | offers a complete set of decision/action alternatives, or |
| Low | 1 | offers no assistance: the human must take all decisions and actions |

Table 1: Levels of automation of decision and action selection.
Reproduced from Parasuraman et al. (2000)

In manual mode, the coordinator retains full authority over every resource allocation decision, functioning at Level 1 of the Parasuraman et al. (2000) scale of decision and action selection automation, where the system offers no assistance and the human takes every decision and action.

In semi-automatic mode, the system proposes a candidate assignment based on automated information analysis and then executes it only once the coordinator actively approves, corresponding to Level 5 of the scale, at which the computer suggests an option and carries it out contingent on human approval.

In fully automatic mode, the system executes assignments independently up to a predefined capacity threshold, corresponding to Level 7 of the scale, where automation executes on its own and then necessarily informs the operator.

Although the scale does not capture it, the coordinator can additionally reverse any assignment after the batch completes, a recovery mechanism that sits outside what the level captures. Rather than fixing a single degree of automation at design time, this switchable architecture embodies the dynamic function allocation logic proposed by Atashfeshan et al. (2021), allowing coordinators to calibrate their level of involvement in real time based on situational demands, task risk, and operational load.

# 3 Methodology

## 3.1 Design Science Research

This thesis is grounded in Design Science Research (DSR), a methodology built around the idea that rigorous academic work can take the form of designing and evaluating artifacts that solve real-world problems (Peffers et al., 2007). Rather than producing explanatory theory, DSR produces things that work, and in information systems this can mean anything from decision support systems to governance strategies and change interventions (Gregor & Hevner, 2013). For this project, that artifact is a functional prototype of a volunteer-task matching and coordination system, which makes DSR a natural fit. The specific process followed here is the six-activity model proposed by Peffers et al. (2007), shown in Figure 2: problem identification, definition of the objectives for a solution, design and development, demonstration, evaluation, and communication.

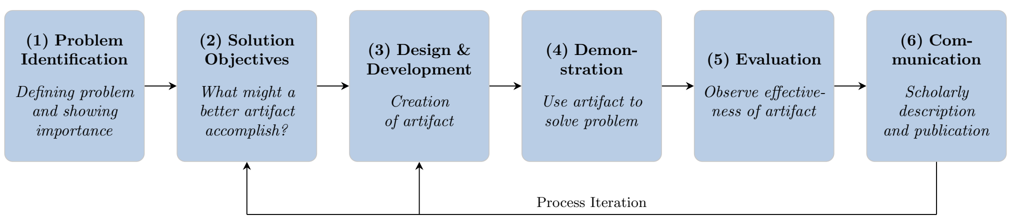


Figure 2: Representation of the DSR methodology process model.
Adapted from Peffers et al. (2007).

The thesis enters the process from a problem-centred starting point, working outward from the coordination failures identified in Section 2. The DSR model permits iteration between stages, but in this project iteration did not take the form of discrete, repeated design cycles; instead, refinement was distributed across the evaluation phase, with changes folded back into the prototype as they surfaced. The individual activities are described in the subsections that follow.

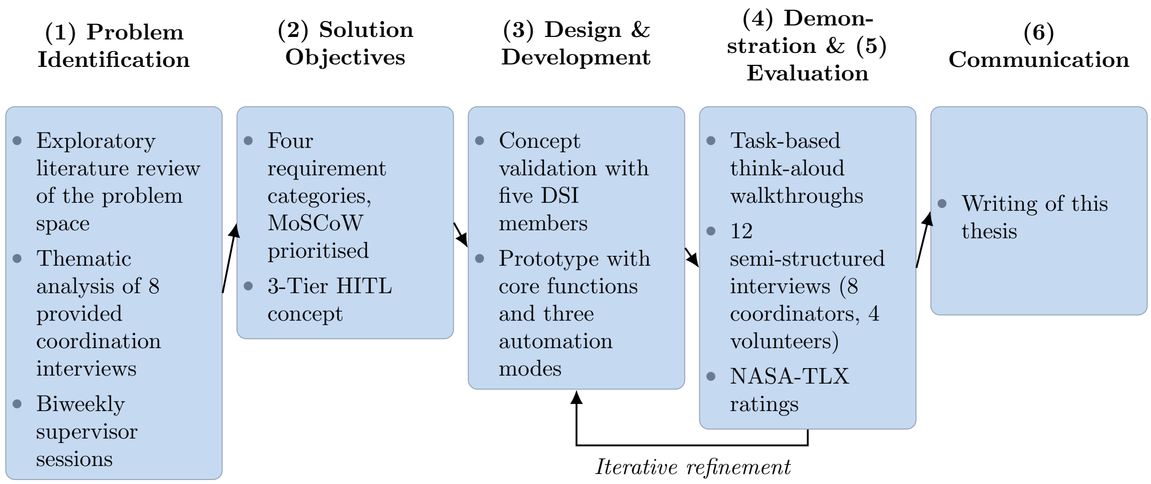


Figure 3: Application of the DSR process model in this thesis.

## 3.2 Requirements Elicitation

This phase corresponds to the first two activities of the DSR process, problem identification and the definition of the objectives for a solution. Requirements for this project were gathered from two sources in parallel. The first was an exploratory literature review (Webster & Watson, 2002), which built an initial understanding of the problem space, covering existing volunteer management systems, crisis coordination challenges, and human-in-the-loop automation concepts. As the picture became clearer, the review became more structured, and recurring themes were consolidated into a first set of requirements that formed the conceptual basis for the 3-Tier HITL approach developed in this thesis.

The second source was a set of pre-existing interviews on volunteer coordination in crisis contexts, provided by the supervising team and listed in Table 2. These were individually listened to, colour-coded, and systematically analysed (Braun & Clarke, 2006) to extract relevant themes and requirements, adding a grounded, practitioner perspective that the literature alone could not fully capture. The elicitation was further supported by biweekly sessions with the responsible supervisor, in which emerging ideas, design directions, and open questions were discussed and progressively refined. Together, these inputs produced the four categories of requirements that defined the prototype concept, presented in full in Appendix A.1. All requirements were prioritised using the MoSCoW method as described in Section 4.3.

| Sphere of Activity | Interviews | Length | Interviewees |
|---|---|---|---|
| **Helpers** | **4** | | |
| Helper 1 | 1 | 1:10:25 | V_Helper_I01 |
| Helper 2 | 1 | 0:45:00 | V_Helper_I02 |
| Helper 3 | 1 | 0:33:32 | V_Helper_I03 |
| Helper 4 | 1 | 0:44:48 | V_Helper_I04 |
| **Hosts** | **4** | | |
| Host 1 | 1 | 0:25:20 | V_Host_I01 |
| Host 2 | 1 | 0:27:00 | V_Host_I02 |
| Host 3 | 1 | 0:32:06 | V_Host_I03 |
| Host 4 | 1 | 0:49:11 | V_Host_I04 |
| **TOTAL** | **8** | **05:27:22** | **8** |

Table 2: List of provided interviews

## 3.3 Validation & Design

This phase covers the design and development activity of the DSR process. Before development began, the proposed 3-Tier HITL concept was subjected to a validation step. It was presented to five members of the Digital Society Initiative at UZH (Katashinskaya et al., 2026), whose critical feedback provided an important external perspective and led to targeted adjustments in both the concept and its scope. Only after these adjustments were incorporated did prototype development start, implementing the core functions and the three automation modes of the system. The design itself was not frozen at this point: as described below, the evaluation phase fed further refinements back into the prototype.

## 3.4 Evaluation

In line with the formative evaluation principles of Sonnenberg & Brocke (2012), who argue that design decisions taken early have a disproportionate effect on the final artifact and that flaws caught late are costly to fix, evaluation was not treated as a final step but distributed across the development process. Once a working version was available, the prototype was evaluated through structured walkthroughs and semi-structured interviews, and the feedback surfacing in these sessions was analysed on an ongoing basis so that requirements and design adjustments could be folded back into the prototype over the course of the interview period.

Each session followed the same three-part structure. It opened with a guided, task-based walkthrough in which participants completed a fixed set of role-specific tasks while thinking aloud, verbalising their actions, expectations, and points of confusion as they worked (Nielsen, 1993). Participants were reminded that the session tested the system and not them, so that any difficulty would be reported as a finding rather than concealed. The walkthrough was followed by the semi-structured interview, and each session closed with a NASA-TLX rating. The role-specific walkthrough steps are detailed below.

Semi-structured interviews were chosen because they combine a prepared set of guiding questions with the flexibility to follow up on unexpected but relevant responses (Myers & Newman, 2007). In total, 12 interviews were conducted, 8 with coordinators and 4 with volunteers, amounting to approximately 8 hours and 40 minutes of recorded material and listed in full in Table 3. The larger share of coordinators reflects that the coordination logic sits at the core of the system's design, while volunteers were included to capture their experience of the matching and application process. Coordinators and volunteers followed separate interview guides adapted to their respective roles, each covering current practice, perceived usability and UI design, cognitive load, and trust in the human-in-the-loop dynamics. Participants were explicitly asked to reflect on their experience relative to tools and systems they already knew, grounding their feedback in real-world practice rather than abstract judgement. Each interview was recorded with participant consent, transcribed, and individually analysed to identify recurring patterns and actionable feedback across participants.

### 3.4.1 Coordinators

The eight coordinator interviews were conducted, one in person and seven remotely via Microsoft Teams, in High German, and lasted between 35 and 67 minutes (see Table 3 for individual durations). After a brief introduction to the project and the prototype, coordinators were optionally shown the volunteer-facing view so that they understood how the two sides of the system relate to one another. Each interviewee was asked to take on the role of a coordinator in a fictitious crisis and to complete the same three-step walkthrough: logging in, creating a task, and managing one task assignment in each of the three automation modes (manual, semi-automatic, automatic). The subsequent interview covered four thematic areas: the coordinator's current organisational system for volunteer coordination, perceived usability and UI design, cognitive load and decision fatigue, and trust dynamics in the Human-in-the-Loop architecture, followed by a general feedback block on the largest risks or missing features they would prioritise for the next iteration. The full interview guide is provided in Appendix A.2.

### 3.4.2 Volunteers

The four volunteer interviews were conducted remotely via Microsoft Teams and lasted between 23 and 33 minutes (see Table 3 for individual durations), following the same template adapted to the volunteer role. After the introduction, each participant took on the role of a volunteer in a fictitious crisis. The walkthrough began with the volunteer registration flow, which requires more profile information than the coordinator login, followed by browsing the available tasks and applying for those of interest; the researcher then accepted or rejected each application from the coordinator side so that participants could observe the consequences of their actions and complete the application loop. The interview covered four thematic areas: prior experience with volunteering during a crisis, perceived usability and UI design, cognitive load during the application process with particular emphasis on the acceptance of the amount of data required at registration, and reflections on past administrative experiences in volunteering contexts. The full interview guide is provided in Appendix A.3.

| **Sphere of Activity** | **Interviews** | **Length** | **Interviewees** |
|---|---|---|---|
| **Coordinator Organisations (CO)** | **8** | | |
| Asyl-Organisation Zürich (AOZ) | 1 | 0:40:50 | CO 1 |
| Schweizerisches Rotes Kreuz (SRK) | 1 | 0:56:17 | CO 2 |
| Direktion für Entwicklung und Zusammenarbeit (DEZA) | 1 | 0:51:13 | CO 3 |
| Reformierte Kirche Kanton Zürich (zhref) | 1 | 1:06:40 | CO 4 |
| Schweizer Tafel | 1 | 0:35:26 | CO 5 |
| Feuerwehr Engstringen | 1 | 0:49:16 | CO 6 |
| Glückskette | 1 | 1:01:10 | CO 7 |
| Reformierte Kirche Kanton Zürich (zhref2) | 1 | 0:50:11 | CO 8 |
| **Volunteers (V)** | **4** | | |
| Volunteer 1 | 1 | 0:27:04 | V 1 |
| Volunteer 2 | 1 | 0:33:29 | V 2 |
| Volunteer 3 | 1 | 0:23:05 | V 3 |
| Volunteer 4 | 1 | 0:25:48 | V 4 |
| **TOTAL** | **12** | **08:40:29** | **12** |

Table 3: List of conducted interviews

After interacting with the prototype, participants also completed a NASA Task Load Index assessment, a standardised and widely used measure of perceived cognitive workload across six dimensions: mental demand, physical demand, temporal demand, performance, effort, and frustration (Cao et al., 2009). Only the rating scale was administered; the pairwise comparison step was omitted, as it offered no meaningful additional value in this context. Ratings were collected relative to the tools and systems participants already used rather than in absolute terms, so the resulting scores express perceived load against existing practice. The scoring and its interpretation are detailed in Section 6.3.

The sixth DSR activity, communication, is served by this thesis as a whole. In practice these activities overlapped rather than running as discrete sequential stages, as the iterative refinement during the evaluation phase shows.

# 4 Problem Identification & Solution Objectives

The prototype addresses two distinct stakeholder groups: the spontaneous volunteer and the coordinator. As established in Section 2.1, this thesis focuses specifically on the coordination of spontaneous volunteers: individuals who offer assistance during a crisis without prior affiliation to a relief organisation and register through the platform to be matched with tasks (FEMA, 2014; Zettl et al., 2017).

## 4.1 Volunteers

Volunteers register on the platform, declare their skills, browse open tasks, and apply for those they are available for. From that point, the coordinator controls the assignment workflow. The volunteer-facing interface is intentionally minimal, surfacing only the information needed for their specific assignment in order to avoid information overload and discourage unsafe self-directed action in the field (Oviatt, 2006; Auferbauer et al., 2016).

## 4.2 Coordinators

The coordinator is the primary actor this thesis targets. Operating on behalf of an organisation or municipality, coordinators are responsible for creating and managing tasks, monitoring deployment status, and controlling the assignment workflow. The 3-Tier HITL system is implemented directly on the coordinator side, giving them explicit control over which automation mode (Manual, Semi-Automatic, or Automatic) is active at any given time. Volunteers experience the consequences of these modes through the assignments they receive, but the delegation logic and its cognitive implications are concentrated in the coordinator role. The specific interaction flows for both roles are detailed in Chapter 5.

## 4.3 Prototype Scope & Requirements prioritisation

The prototype was scoped with a deliberate focus on usability and the demonstration of the 3-Tier HITL concept. Features such as complex security mechanisms and mathematically optimised matching algorithms for the semi-automatic & automatic mode were excluded, as their inclusion would not have meaningfully contributed to evaluating the core innovative contribution within the constraints of this thesis. This scoping decision directly shaped how requirements were elicited and subsequently prioritised.

To structure the prioritisation process, the MoSCoW method was applied. MoSCoW is an acronym standing for Must-have, Should-have, Could-have, and Won't-have, and provides a structured framework for categorising requirements according to their significance and urgency (Vijayakumar et al., 2024). Must-have requirements represent specifications that are vital to the project's success and must be fulfilled for the system to function at all. The Should-have requirements are significant but not critical for immediate delivery, adding value once the essential features are in place. Could-have requirements are desirable extras that can be included if time and resources allow, but are not prioritised over the two higher categories. Won't-have requirements are explicitly excluded from the current scope, either deferred to a later iteration or deemed low priority for this development cycle. The method was chosen because of its simplicity and its ability to support transparent scope management, allowing clear trade-off decisions to be communicated between stakeholders (Vijayakumar et al., 2024). The complete set of prioritised requirements is presented in Appendix A.1.

## 4.4 Requirements

The requirements elicited from literature and interviews are organised into four categories. Functional Requirements (FR) define what the system must do, Non-Functional Requirements (NFR) constrain system-level qualities such as performance and security, Information Requirements (IR) specify the data the system must capture and store, and Design Requirements (DR) govern interface-level design decisions derived from cognitive load theory. All requirements were prioritised using the MoSCoW method as described in Section 4.3.

### 4.4.1 Functional Requirements

Functional requirements define the observable behaviours the system must support, specifying what the system shall do in response to user actions or system events ("ISO/IEC/IEEE International Standard - Systems and software engineering – Life cycle processes – Requirements engineering", 2018). The 31 requirements listed in Table 5 fall into seven groups: volunteer registration and onboarding (FR-01 to FR-05), task creation and management (FR-06 to FR-07), the assignment pipeline and its automation modes (FR-08 to FR-11), matching constraints and workload balancing (FR-12 to FR-17), volunteer-facing information and communication (FR-18 to FR-23), records, screening, and coordinator support (FR-24 to FR-27), and a set of operational refinements grounded in interview data (FR-28 to FR-31). These reflect two recurring tensions in volunteer coordination: the need to reduce coordinator workload through automation, and the need to preserve volunteer agency over the assignment process.

The first tension is most visible in FR-09 through FR-11, which together define the system's assignment pipeline. FR-09 mandates three distinct automation modes (manual, semi-automatic, and automatic). FR-10 adds a coordinator review gate in semi-automatic mode, ensuring proposals are inspected before volunteers are notified. FR-11 cuts across all three modes with a hard constraint that volunteer agency shall be preserved throughout the assignment process, meaning volunteers must apply for open tasks themselves, and no assignment is confirmed without an initial act of self-selection on the volunteer's part.

A second cluster of requirements addresses operational friction that the literature had not adequately captured. FR-28 through FR-31 emerged directly from interview data and encode four concrete pain points: the cognitive burden of manually monitoring shift headcounts (FR-28, FR-30), the repetitive effort of recreating identical weekly schedules from scratch (FR-29), the frustration caused by already-filled tasks remaining visible and generating double-bookings (FR-30), and the information overload produced by unfiltered chat-based coordination (FR-31, FR-19).

### 4.4.2 Non-Functional Requirements

Non-functional requirements define the quality constraints under which the system must operate, specifying how well it performs rather than what it does ("ISO/IEC/IEEE International Standard - Systems and software engineering – Life cycle processes – Requirements engineering", 2018). The 15 requirements listed in Table 6 reflect the operational realities of crisis coordination. Access must be fast and reliable under surge conditions (NFR-02, NFR-03), and information exposure to volunteers must be carefully controlled to prevent unsafe self-directed behaviour (NFR-12). NFR-01 addresses the coordinator interface directly, requiring that task status be visualisable at a glance without cross-referencing external lists. NFR-10, which calls for brief match explanations alongside automated suggestions, sits at the boundary between non-functional and design concerns: it does not change what the system computes, but it directly shapes how much coordinators can trust and audit the system's reasoning. NFR-14 and NFR-15 emerged from interview data, encoding the need for automated vetting of incoming volunteers and a single authoritative information source to prevent the rumour propagation that coordinators described as a persistent operational hazard.

### 4.4.3 Information Requirements

Information requirements specify what data the system must store and maintain to support its functions (Davis, 1982). The 14 requirements listed in Table 7 cover three layers: volunteer data (IR-01 through IR-06), task data (IR-08 through IR-11), and operational records (IR-12 through IR-14). Together they define a deliberately minimal schema, reflecting the data minimisation principle encoded in NFR-07: only the fields genuinely needed for matching, coordination, and post-incident reporting are captured. IR-07, which mandates a structured assignment status vocabulary (proposed, accepted, declined, completed, cancelled), is the information-layer counterpart to the behavioural pipeline defined by FR-09 through FR-11.

### 4.4.4 Design Requirements

Design requirements specify how the user interface shall be structured to minimise coordinator cognitive load, translating the cognitive load and HCI principles reviewed in Section 2.4.1 into concrete interface guidelines (Brace & Thramboulidis, 2010). The 18 requirements listed in Table 8 are organised into eight categories, each targeting a distinct cognitive mechanism: limiting working memory demands (DR-01–DR-02), eliminating extraneous load (DR-03–DR-06), reducing decision complexity via Hick's Law (DR-07–DR-09), supporting existing mental models (DR-10–DR-11), leveraging multimodal presentation (DR-12–DR-13), adapting to user expertise (DR-14–DR-15), supporting representational needs (DR-16), and applying usability as a load-reducing mechanism (DR-17–DR-18). Taken together, they operationalise the argument that cognitive load is not only an instructional concern but a direct constraint on interface design in high-pressure coordination environments. How each requirement was applied in practice is discussed in Section 5.4.

# 5 Design & Development

## 5.1 System Overview

At its core, the prototype is a web-based volunteer-task matching and coordination system that digitalises the concept of a Volunteer Reception Center (FEMA, 2014). Rather than managing spontaneous volunteers through fragmented spreadsheets and messaging channels, coordinators interact with a single dashboard that consolidates registration, task management, and assignment workflows into one interface. The system was built on the requirements established in Section 4.4, with targeted refinements incorporated following the evaluation interviews documented in Section 6.4. The following sections detail the data model, the three delegation tiers, the interface design, a recorded walkthrough of key screens, and the technical architecture of the deployed system.

## 5.2 Data Model

The data model underlying CrisisMatch consists of five entities, shown in Figure 4. Rather than modelling a fully-featured user management system, the schema is shaped around one question: what does the coordination process actually need at runtime?

The `Volunteer` entity reflects a deliberate choice to collect as little as possible. Volunteers in crisis settings are often spontaneous participants who have no prior relationship with the coordinating organisation, and asking too much up front creates friction and erodes trust. Keeping registration lean is therefore not only a privacy concern but a practical usability requirement: the more a system demands, the fewer people complete the process, particularly under crisis conditions (Betke, 2018). As one coordinator confirmed, volunteers are simply less willing to sign up when they feel the process is intrusive (CO 6, 25:50).

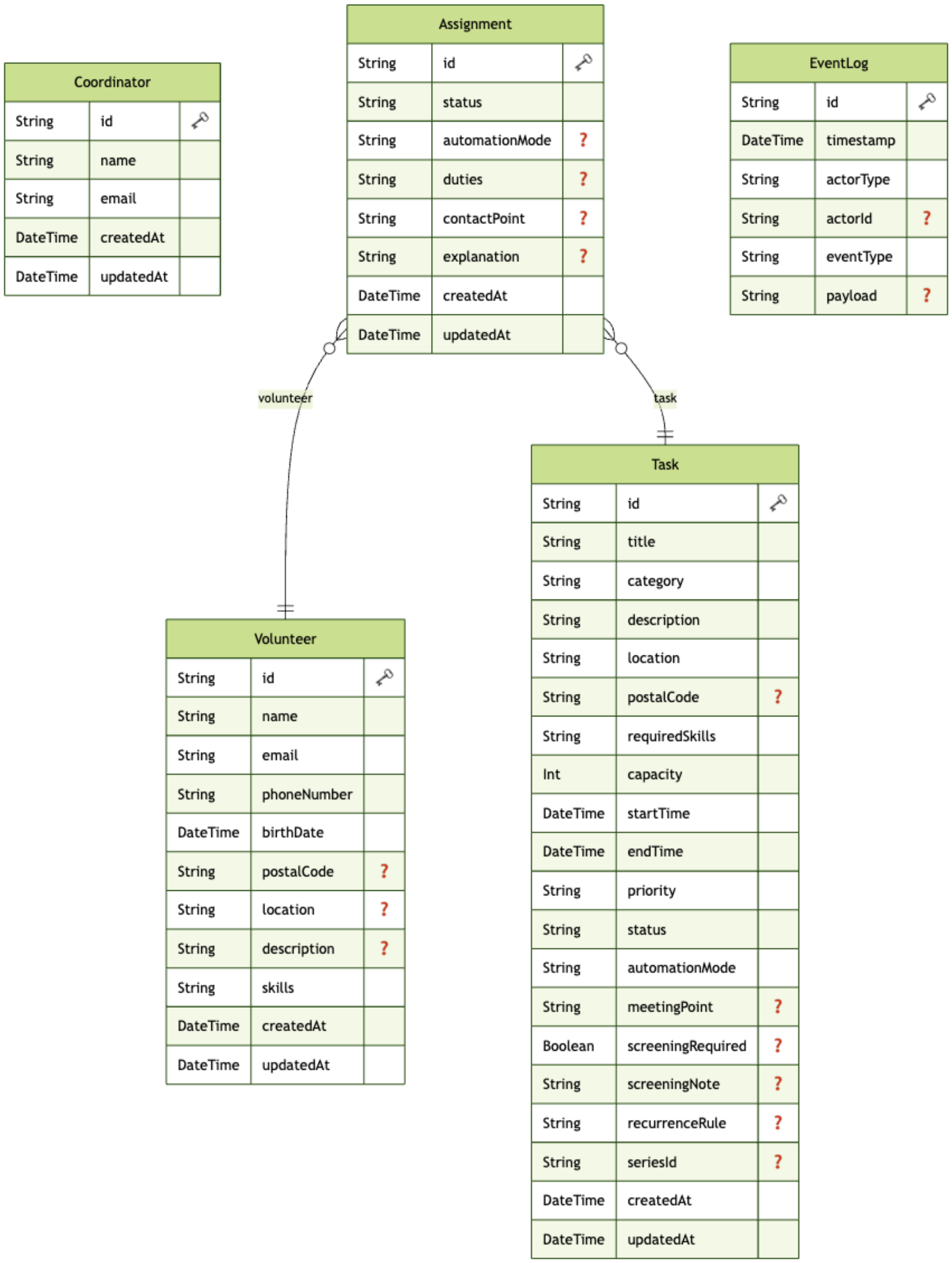


Figure 4: Entity-Relationship-Model CrisisMatch

Drawing this line is not straightforward, however. Too much restraint in data collection risks degrading the quality and safety of assignments, since a coordinator matching volunteers to tasks with sensitive or physically demanding requirements depends on having reliable profile information. The profile as modelled here therefore covers only what matching and contact require, navigating this trade-off in favour of accessibility and in line with NFR-07.

The `Task` entity is where a coordinator describes what help is needed and, critically, how much of the matching process the system should handle on its own. The automationMode field is the concrete anchor point for this decision in the data model: it determines, per task, whether the system acts as a suggestion engine or as an autonomous dispatcher. Supporting fields for recurrence and screening allow the entity to handle the operational diversity of real deployments without compromising the core structure.

The `Assignment` entity connects a volunteer to a task and records the outcome of the matching process. Beyond status tracking, it carries an explanation field that makes the system's reasoning visible to the coordinator, which is the practical expression of NFR-10. Coordinators can also attach role-specific instructions directly to an assignment, keeping relevant context in one place.

The `Coordinator` entity is intentionally thin, since the coordinator's role in the system is expressed through actions rather than stored attributes. The `EventLog` captures those actions, writing a timestamped record of every significant event in the system regardless of whether it was triggered by a human or by the automation. This makes it possible to reconstruct what happened during any deployment after the fact, satisfying NFR-04 without adding any weight to the operational flow.

## 5.3 3-Tier Human in the Loop

The central design decision in CrisisMatch is that the degree of automation is not a global system property but a per-task configuration. Each task carries an automationMode field set at creation time, placing it into one of three tiers (FR-09). This directly determines which interface elements are rendered and which API paths are triggered when a coordinator interacts with an assignment.

### 5.3.1 Manual

In Manual mode, the system does not invoke the matching engine at all, placing full allocation authority with the coordinator in line with Level 1 of the Parasuraman et al. (2000) model at the decision and action selection stage (see Table 1). When a volunteer applies, their assignment is recorded with status PROPOSED and surfaced in the task detail view alongside a warning note prompting the coordinator to conduct offline vetting before accepting (FR-26). The interface provides a pre-composed email template, shown in Figure 5, that opens the coordinator's mail client with the volunteer's address and a contextually populated message body, enabling direct contact through existing email tools without building a dedicated in-app messaging system (FR-23).

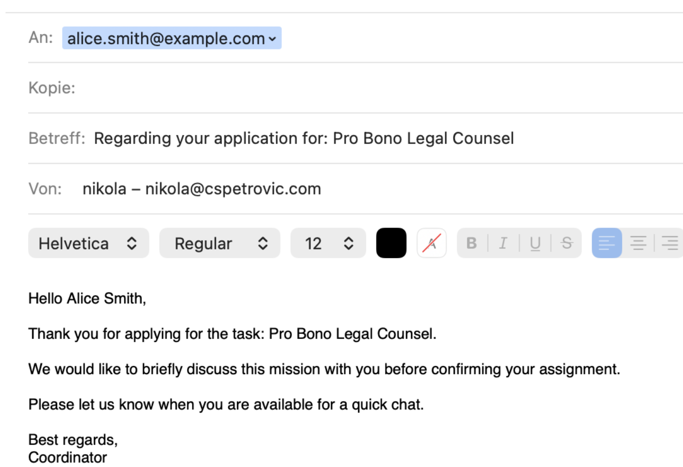


Figure 5: Pre-composed Mail Template

Accept and Decline buttons are available directly on each applicant card, but no scoring or ranking information is shown. A separate search panel allows coordinators to look up volunteers by name, email, or postal code and assign them manually (FR-27); all capacity and uniqueness constraints are still enforced server-side regardless of how the assignment originates (FR-14, FR-30). This mode is intended for tasks where the coordinator must exercise personal judgement, typically high-priority or sensitive roles, and the system's role is purely administrative (Betke et al., 2023).

### 5.3.2 Semi-Automatic

In Semi-Automatic mode, the coordinator retains the final decision but can request a scored ranking of current applicants on demand. The system proposes ranked candidates and executes each assignment only once the coordinator approves it, corresponding to Level 5 of the Parasuraman et al. (2000) scale, where the computer carries out a suggestion contingent on human approval, with the coordinator's authority over every individual acceptance preserved (Onnasch et al., 2014). Pressing the `Score & Rank Applicants` button triggers a POST `/api/tasks/[id]` call with action suggest (FR-08, FR-10). The server queries all assignments with status `PROPOSED` for that task, runs each applicant's profile through the scoring function in `lib/matching.ts`, and returns the results sorted by descending score. The scoring function combines three components: a skill match score (the count of required skills the volunteer holds, or 1 if the task has no requirements) (FR-12), a location score (2 for an exact postal code match, 1 for a prefix match) (IR-06), and a priority boost (0–3 depending on the task's priority level), weighted as:

$$\text{score} = \text{skillScore} \times 3 + \text{locationScore} \times 2 + \text{priorityBoost}.$$

Skills are weighted most heavily because skill mismatch is the primary safety concern in volunteer deployment (FEMA, 2014); location is secondary, reflecting the operational benefit of proximity in real-world dispatch (Manshadi & Rodilitz, 2022); and priority acts as a tiebreaker to direct volunteers toward the most urgent tasks first. In the current scope the priorityBoost is irrelevant, since all volunteers are evaluated inside the same task and therefore receive the same boost. It was added purely for the possibility of future multi-task assignments, which are out of scope for this prototype. Alongside each ranked candidate the interface displays an explanation string stating the skill match score and whether a location proximity match was found, as seen in Figure 7 (NFR-10). The coordinator then accepts or declines individual suggestions. Each action calls PATCH `/api/assignments/[id]`, and the suggestion is removed from the candidate list once the coordinator has acted on it.

### 5.3.3 Automatic

In Automatic mode, the coordinator acts in a supervisory rather than an approving role, corresponding to Level 7 of the Parasuraman et al. (2000) scale, where the system executes automatically and then necessarily informs the operator. Applicants appear in the assignment list with a `Waiting for auto-fill...` indicator, the coordinator can remove any individual applicant before processing begins. When the coordinator is satisfied with the applicant pool, pressing `Run Auto-fill Batch` triggers a POST `/api/tasks/[id]` call with action autoFill (FR-08, FR-09). The server scores all `PROPOSED` applicants using the same function as Semi-Automatic mode, sorts them by score, and accepts the highest-ranked candidates up to the remaining capacity within a single database transaction (FR-28). Task status is recalculated atomically after the batch completes, and a post-action summary is displayed to the coordinator stating how many volunteers were assigned (FR-30, NFR-01).

This mode deliberately avoids fully passive automation: the coordinator must still trigger the batch, preserving a moment of intentional human engagement before the decision is executed. Beyond that trigger point, the coordinator retains the ability to remove already-assigned volunteers at any time via an explicit Remove button, allowing corrections when circumstances change after the batch has run, for instance if a volunteer becomes unavailable or the task requirements shift (FR-07, NFR-01). This design directly addresses the out-of-the-loop unfamiliarity problem identified by Parasuraman et al. (2000) and the sharp increase in situational awareness loss documented by Onnasch et al. (2014) when automation crosses from information analysis into action selection without human confirmation. Rather than fixing a single degree of automation at design time, this per-task switchable architecture reflects the principle of dynamic function allocation described by Atashfeshan et al. (2021), in which automation levels are not locked in during system design but can be adjusted based on operational conditions and task requirements. The explanation field on each resulting assignment records the system's reasoning, making every automated decision inspectable after the fact (NFR-10, IR-14).

## 5.4 UI/UX Design

The interface design of CrisisMatch was guided throughout by a set of evidence-based design requirements (DR-01–DR-18) derived from Cognitive Load Theory and established HCI principles, listed in full in Section A.1. Rather than treating these as a post-hoc evaluation rubric, they functioned as active constraints during construction, shaping decisions about layout, component structure, interaction patterns, and visual language. What follows is an account of those decisions and the reasoning behind them.

### 5.4.1 Information Architecture and Progressive Disclosure

As shown in Figure 6, the dashboard is organised around a deliberately flat two-level hierarchy: a task list at the overview level, and a dedicated detail page for everything else. This structure was chosen to keep the coordinator's navigational context shallow. At any moment, the coordinator is either scanning the list of active tasks or working within a single task, never three levels deep, never required to hold the contents of one screen in mind while reading another.

This directly satisfies the recognition-over-recall principle (DR-11): the system externalises state rather than asking the user to remember it. On the task list itself, each card exposes only the information that is necessary for a coordinator to decide whether to open it: the task title, location, date, automation mode, and a capacity progress bar. Fields such as description, skill requirements, and meeting point are intentionally withheld from the list view and surfaced only on the detail page. This is

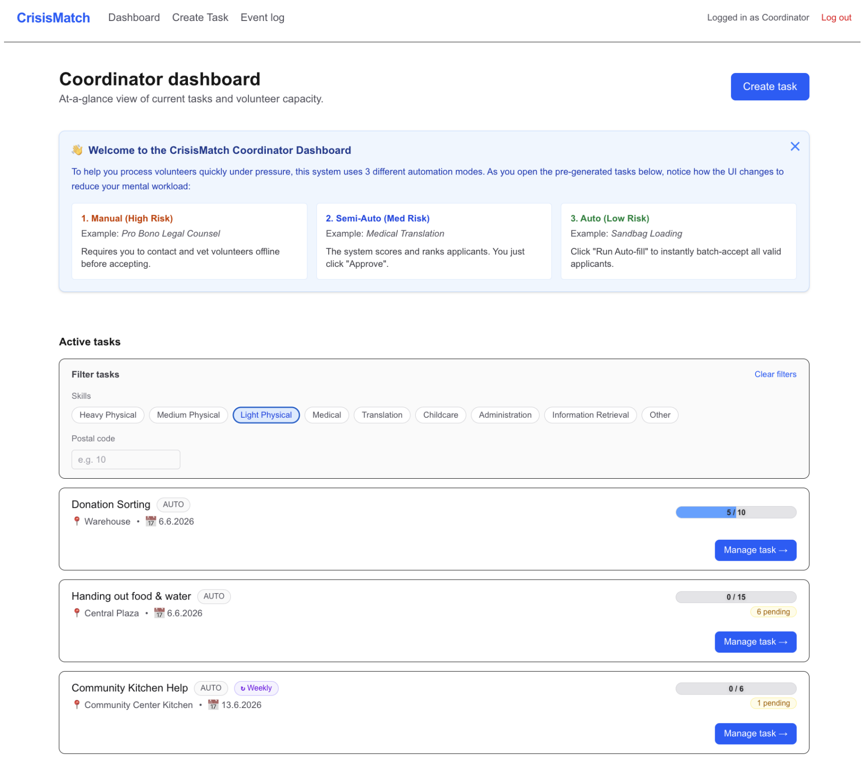


Figure 6: Coordinator Dashboard

not an arbitrary simplification, it reflects DR-01, which holds that working memory can comfortably process only three to four novel information elements simultaneously, and DR-08's principle of progressive disclosure, presenting the overview first and revealing detail on demand. The task detail page reinforces spatial integration (DR-03, the Split-Attention Principle) by keeping the task header persistently visible at the top of the page `title / location / time window / priority / automation mode` while the coordinator browses applicants below. There is no need to navigate away to recall what skills or location the task requires, since that context is always present in the same view.

### 5.4.2 Volunteer Profile Cards and Chunking

Each volunteer in the matching workflow is represented by a `VolunteerProfileCard` component, displayed in Figure 7. The card was designed explicitly around DR-02, which argues that users process pre-organised informational schemas as single units, reducing working memory load. Rather than presenting a flat list of fields, the card groups information into three spatial chunks: `identity` (name, email, and a status badge where applicable), `logistics` and `context` (postal code, system match explanation), and `actions`, which vary by mode and assignment state (accept, decline, contact, remove). Skills and free-text descriptions are hidden by default behind a `View details` toggle, keeping the card compact for coordinators who do not need them and accessible for those who do, again following DR-08 and DR-07's instruction to limit the number of active decision alternatives at any moment. When the matching engine has scored a volunteer, its reasoning is surfaced directly on the card as a highlighted `System Match` note, co-located with the volunteer's identity rather than placed in a separate panel.

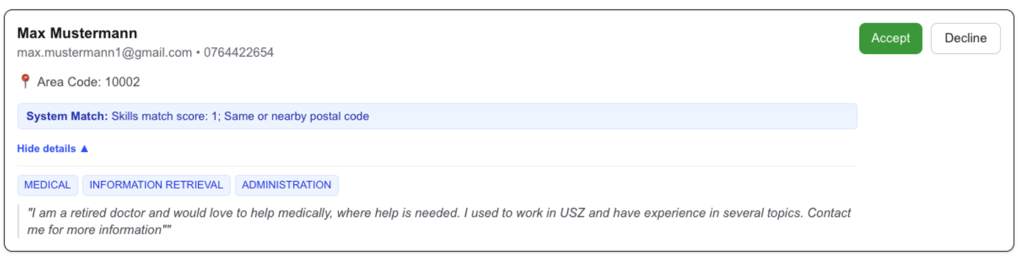


Figure 7: Volunteer Profile Card after ranking in semi-automatic mode

This is a direct application of DR-03: if the coordinator must integrate a score with a name and a skill list, those elements should appear together, not in separate locations that require the eye to travel. Volunteers who have submitted an application while it remains under review may retract it via a `Withdraw application` link rendered beneath each pending entry on the dashboard. The action updates the assignment status to `CANCELLED` (IR-07) and removes the entry from the pending list immediately.

Withdrawal is restricted to the `PROPOSED` state, giving concrete expression to FR-11: since no assignment should proceed without an initial act of voluntary self-selection, a volunteer must be equally free to retract that self-selection while it remains unconfirmed.

### 5.4.3 Reducing Decision Complexity

Several design choices in the system were motivated by Hick's Law (DR-07), which states that decision time increases logarithmically with the number of presented alternatives. In Semi-Automatic mode, the scoring engine pre-ranks all applicants and presents only the candidates needed to fill remaining capacity, not the full applicant pool (DR-09). Skill selection uses a grid of toggle buttons drawn from a fixed vocabulary rather than a free-text field (DR-18), eliminating spelling errors and reducing the cognitive cost of formulating input. Priority, automation mode, and recurrence are all constrained to dropdown or radio selection for the same reason. The principle throughout is that wherever a closed set of valid options exists, the interface should offer it as a structured choice rather than asking the coordinator to construct an answer from scratch.

The same logic extends to task discovery. In an active deployment the task list can grow to a length where scanning it in full imposes a recognition burden that compounds under time pressure (DR-01). A shared `TaskFilterBar` component, rendered on the volunteer dashboard and the coordinator task list alike, lets users narrow the visible set by required skills and postal code prefix (FR-31, NFR-01). Skill selection uses the same toggle-button vocabulary established elsewhere in the interface (DR-18), introducing no new interaction pattern. A `Show tasks matching my skills` shortcut on the volunteer side pre-populates the filter from the volunteer's own registered profile in a single tap, collapsing the most common filtering intent to minimal input (DR-07). The full volunteer dashboard is shown in Figure 8.

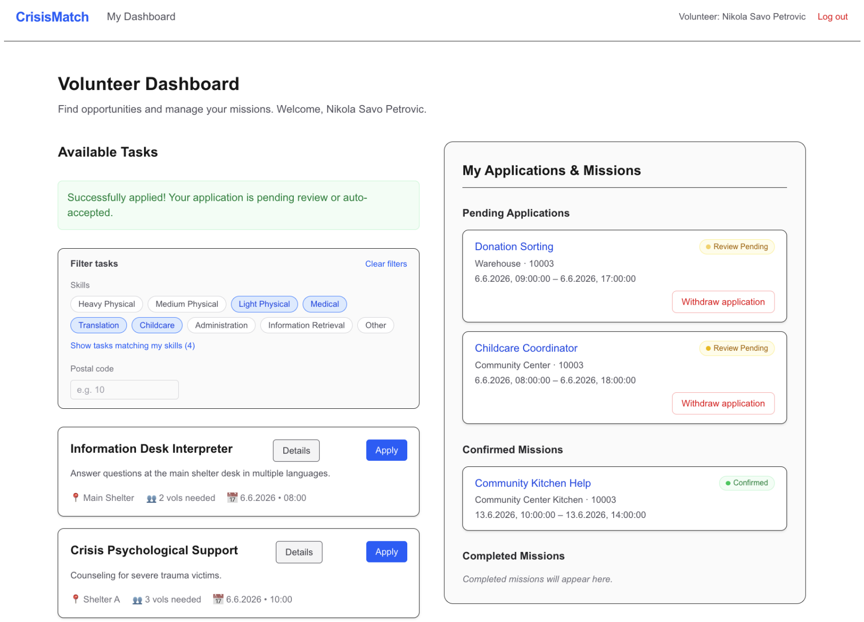


Figure 8: Volunteer Dashboard

### 5.4.4 Onboarding and Expertise Adaptation

A dismissible onboarding banner appears on the coordinator dashboard at first login and is suppressed on subsequent visits via `localStorage`, implementing the guidance-fading principle (DR-14): scaffolding useful to a first-time user becomes redundant once the coordinator is familiar with the system. Following the worked-example effect (DR-15), the banner illustrates each automation mode with a concrete case: a high-risk legal task requiring offline vetting, a medium-risk translation task where the coordinator approves the top-scored applicant, and a low-risk labour task suited to batch auto-fill, rather than describing the modes abstractly.

### 5.4.5 The Automation Mode Selector

The `AutomationModeEvaluator` component on the task creation form is perhaps the interface element that most directly embodies the thesis's central design tension: automation should be configurable without placing undue cognitive burden on the person doing the configuring. The component offers two distinct paths. The default `Choose directly` path renders the three modes as side-by-side radio cards, each with a one-line tagline and a short explanatory sentence. A coordinator who already knows the risk profile of their task can make the selection in two clicks, with no additional friction. The `Guide me through selection` path activates a structured two-step wizard: the coordinator rates four risk factors, physical danger, legal liability, data sensitivity, and certification requirements, on a three-point scale, then selects an urgency level as shown in Figure 10.

A decision matrix maps these inputs to a recommendation in real time, presenting the outcome as a live risk score, a colour-coded risk badge, and a named mode recommendation with an explanation. One deliberate feature of the guided path is a hard override: when certification risk is rated high, the recommendation is locked to Manual regardless of the other factor scores. This externalises a safety constraint that would otherwise require the coordinator to reason about it themselves, reducing both cognitive load and the risk of misconfiguration. The full 3x3 risk matrix is available as a collapsed disclosure element for coordinators who wish to inspect the logic, satisfying DR-08's instruction to reveal detail on demand rather than presenting it by default (see Figure 9).

▼ View full decision matrix

| Urgency \ Risk | LOW Risk | MEDIUM Risk | HIGH Risk |
|---|---|---|---|
| LOW | Mode 3 — Auto | Mode 2 — Semi-Auto | Mode 2 — Semi-Auto |
| MEDIUM | Mode 3 — Auto | **Mode 2 — Semi-Auto** | Mode 1 — Manual |
| HIGH | Mode 2 — Semi-Auto | Mode 1 — Manual | Mode 1 — Manual |

Figure 9: Risk Matrix

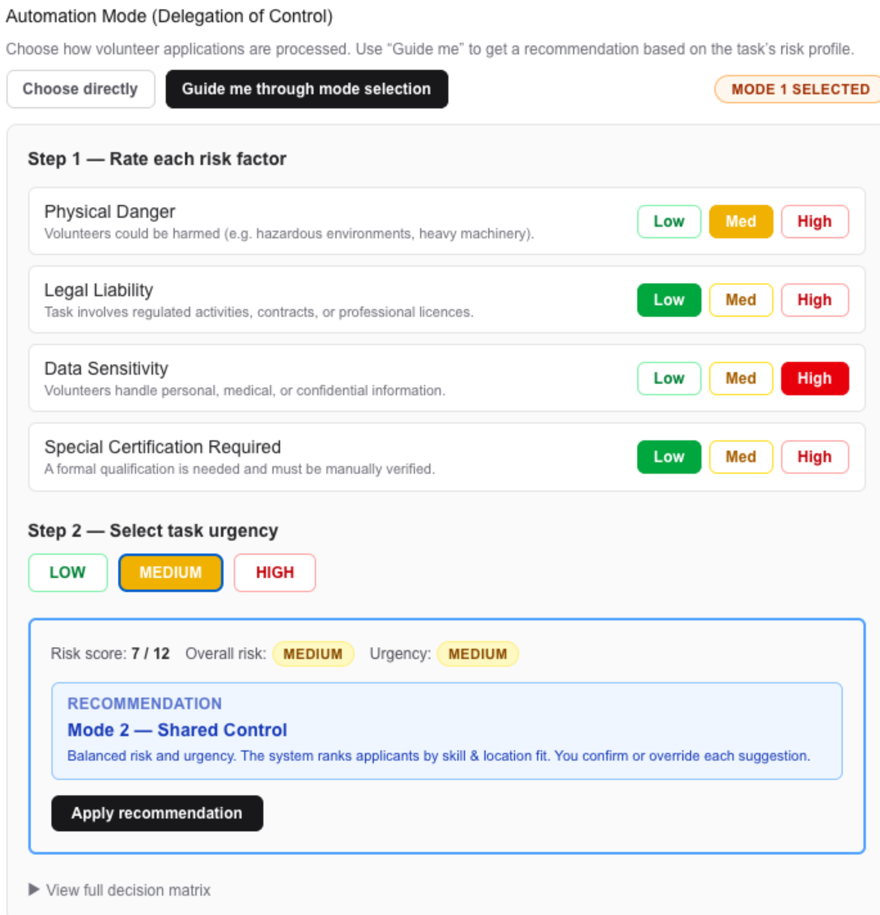


Figure 10: Automation Wizard

The two-path structure is a direct implementation of DR-14. A novice coordinator benefits from the guided evaluator's structured reasoning support, while an experienced one is not burdened by it. Both paths converge on the same stored value and the same downstream matching behaviour.

### 5.4.6 Visual Language

The visual design uses a deliberately restricted and consistent colour vocabulary. Zinc grays carry structural and neutral elements; blue marks primary actions and system-generated information; green signals confirmed assignments and low-risk states; orange indicates manual mode warnings; red surfaces critical-priority tasks and destructive actions.

This mapping is applied consistently across every surface in the application. The consistency is not aesthetic but functional: a coordinator who has learned what blue means on the suggestion panel does not need to re-learn it on the task list. Each encounter with a coloured element requires less interpretation than it would in a less disciplined system, reducing extraneous load over time (DR-17, learnability and memorability as load-reducing mechanisms). Navigation is kept intentionally minimal. The application header provides links to the dashboard, the task creation form, and the event log. Within task pages, a `Back to Dashboard` link handles upward navigation. There are no floating panels, modals, or unsolicited notification banners anywhere in the application, in keeping with DR-05 (minimise extraneous interface features) and DR-06 (eliminate interruptions during active matching tasks). The event log, though accessible from the header, is entirely absent from the task management flow; it serves as an after-the-fact auditing surface and imposes no cognitive cost on coordinators who are not actively using it.

## 5.5 Walkthrough

A full walkthrough of the prototype is available as a video (YouTube, or downloadable version), and the live application can be accessed directly at this link. As it is hosted on Render's free tier, the server may take a minute to start up on first access.

## 5.6 Tech Stack & Architecture

CrisisMatch is a full-stack web application built with Next.js 16 (App Router), React 19, and TypeScript 5 in strict mode. Next.js was chosen for its co-location of server and client code: API routes and page components share a single repository and the same types, removing the need for a separately maintained backend. The data layer uses Prisma 6 as the ORM against a SQLite database, chosen for its zero-configuration setup in a prototype context, where the entire database is a single file that can be seeded and reset in seconds (`npx tsx prisma/seed.ts`).

As SQLite lacks native enum and array types, both are stored as strings and JSON-stringified arrays and cast back at the application layer, with Prisma's generated client preserving type safety throughout. The frontend is styled with Tailwind CSS 4 applied directly in JSX, with no component library: all elements, from cards and badges to the risk evaluator wizard, are built from Tailwind primitives, keeping the dependency surface small and giving precise control over the visual language described in Section 5.4.6. Authentication is handled client-side via `localStorage` and a React context (`AuthContext`), with no server sessions, JWTs, or cookies. This is a deliberate simplification: the prototype targets demonstration and evaluation rather than production, so removing the server-side auth layer cuts infrastructure complexity without affecting the research questions.

The matching logic is isolated in a single module (`lib/matching.ts`) of pure functions independent of the HTTP layer, so the scoring algorithm could be tested or replaced without touching the route handlers. API routes follow a consistent pattern: each wraps its logic in a `try/catch`, uses `prisma.$transaction` for atomic multi-writes, and returns typed `NextResponse.json` responses. The seed script creates a deterministic dataset of sixteen volunteers across eight skill profiles and tasks spanning all three automation modes, sufficient to demonstrate every workflow path without manual setup. Figure 11 summarises how these components connect across the client, server, and data tiers.

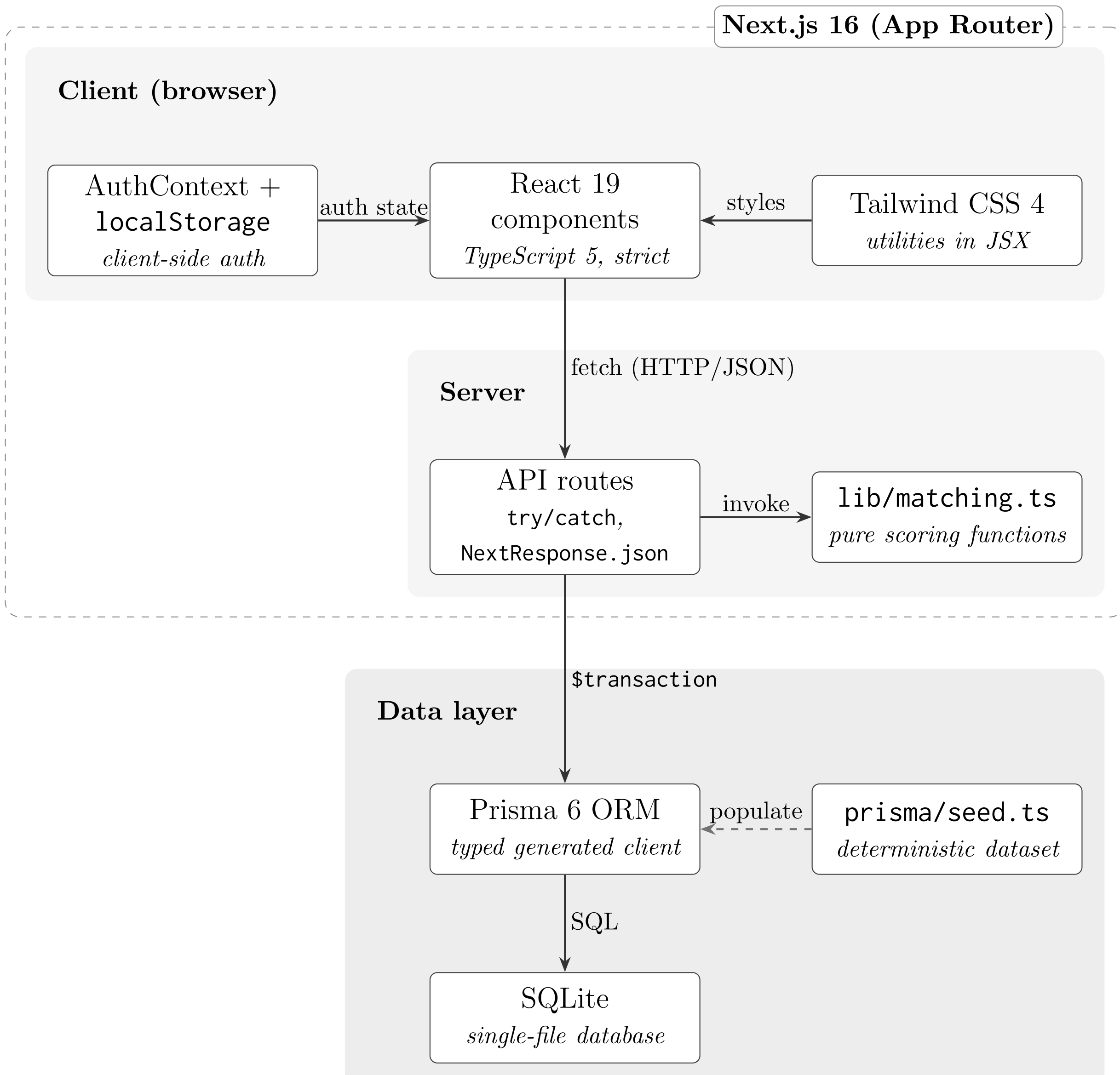


Figure 11: Architecture and tool interconnections of the CrisisMatch prototype

# 6 Evaluation

The evaluation draws on a broad empirical foundation: 12 interviews totalling approximately 8 hours and 40 minutes of recorded material (see Table 3 for durations and codes). It set out to assess three things: usability, perceived reduction in cognitive load relative to current tools, and practical acceptance of the three-tier Human-in-the-Loop architecture.

## 6.1 Coordinators

The findings below are organised by the four thematic areas of the coordinator interview guide (see Section 3.4). Direct quotations are given in the original German; interviewee codes follow the scheme in Table 3.

### 6.1.1 Current organisational systems and time pressure

Without exception, the participating organisations described their current coordination as low-automation: none used algorithmic matching, relying instead on manual matching, spreadsheets, email, or simple scheduling tools (e.g. Doodle). DEZA and Feuerwehr Engstringen do not rely on spontaneous volunteers, hence the comparison to their current system must be treated slightly differently. AOZ matches volunteer profiles by hand (CO 1), the SRK still relies on "klassische Excel-Tabellen" over email (CO 2, 33:34), Schweizer Tafel routes applications through a central inbox administered "mit viel personellem Aufwand" before forwarding them to regional teams (CO 5, 22:59), and Glückskette runs one Doodle poll per shift drawn from a central volunteer database (CO 7, 34:48). CO 8, a second coordinator from zhref, described an in-house tool fed by an obligatory intake interview, combined with a collective email to all volunteers plus selective individual outreach to those she knows to be reliable (CO 8, 33:51). This low-automation baseline provides a clean reference point against which the prototype's automation was assessed.

More significant, however, was a convergent challenge to the thesis's central premise that coordinators routinely operate under acute time pressure. Three coordinators stated that, for steady-state volunteering, time pressure is not the binding constraint. CO 5 was explicit, stating “Wir haben in diesem Bereich keinen Zeitdruck” (CO 5, 25:09), and instead described a surplus of applicants that must be filtered against a hard requirement. CO 4, a domain expert who trains volunteer coordinators across congregations, argued more strongly that designing around urgency is counterproductive for ordinary volunteering: “Zeitdruck ist nie gut [...] nie Freiwillige unter Druck setzen [...] dann verschrecken sie” (CO 4, 49:51). CO 1 similarly located AOZ’s speed pressure in everyday one-to-one matching rather than crises. The premise was, however, confirmed at the high-volume, event, and crisis-windowed end of the spectrum. CO 6, who coordinated large-scale events of up to 1,600 people, described the time pressure simply as “Massiv” (CO 6, 28:35), driven by no-shows and constant replanning; CO 3 (DEZA) confirmed extreme but structurally absorbed time pressure in professional disaster deployment; and CO 7 located it in a tight organisational window, with her colleague and herself having “vielleicht nochmals 5 Tage Zeit, um alles zu organisieren” (CO 7, 36:39) once a collection day is approved. CO 8 added a third variant, describing not an episodic crisis window but a chronic, structural overload across her whole peer group: “die Zeit ist immer ein Problem [...] meine Kolleginnen sagen alle genau das Gleiche” (CO 8, 35:17). The evidence therefore suggests the time-pressure framing holds for crisis, large-event, and crisis-windowed activities, and as a chronic workload condition, but does not generalise to routine steady-state volunteer management. This limits the premise, not the tool: even without time pressure, steady-state coordinators still face a matching problem (CO 4, CO 5), so the system is better justified by the matching effort it removes than by the speed it adds.

A second cross-cutting theme was that the binding constraint for several organisations sits upstream of coordinator decision load, in the quality and screening of volunteer data. The SRK gave the most pointed example: during the Ukraine response they held 2,500 volunteers but could not deploy the right fifteen because certificate information was not documented (CO 2). CO 7 reinforced this from a different angle, noting that volunteer-side data errors propagate to the end user: “wenn ich jetzt meinen Namen falsch schreibe, dann kommt das beim Spender falsch an" (CO 7, 39:02). For these organisations, the matching value proposition is real but gated by input data the system does not itself control.

### 6.1.2 Perceived usability and UI design

Usability feedback was strongly positive across the set. Coordinators described the prototype as "sehr angenehm, sehr übersichtlich" (CO 1, 25:55), "sehr intuitiv [...] sehr einfach zu bedienen" (CO 3, 34:35), "grundsätzlich praktisch und gut [...] logisch aufgebaut" (CO 4, 48:41), and "relativ einfach zu gebrauchen [...] selbsterklärend" (CO 7, 43:42). CO 2, whose reference system is the shift-planning tool Staffomatic, gave the strongest comparative verdict, judging the prototype "zugänglicher und [...] intuitiver gemacht" (CO 2, 55:27) than his existing tool, whose coordinator view he described as cluttered. CO 8 praised the visual aesthetic, expressing surprise at how "hell und leicht" the interface was and adding "das würde ich so gerne so lassen" (CO 8, 38:48). Several participants reported orienting themselves within the interface very quickly, though most explicitly noted that the session was guided and that fluency would require a few repetitions (CO 2, CO 3, CO 4, CO 8).

The rationale for the three-tier automation model was self-evident to most coordinators, who read it as driven by task complexity or content (CO 2, CO 4, CO 5, CO 7, CO 8). CO 8 judged the distinction "sehr logisch" (CO 8, 37:31). One nuance is worth recording. CO 1 found the visual layout immediately clear but only grasped why three tiers existed during the guided walkthrough, stating "während dem Tutorial hab ich es dann verstanden" (CO 1, 26:27). This suggests that visual clarity and conceptual self-evidence are distinct, and that the purpose of the tiers benefits from explicit onboarding.

A number of concrete, actionable UI issues recurred. The most convergent concerned colour: CO 4 (from a design background) and CO 5 requested stronger colour coding of states and risk levels, while CO 7 identified a genuine defect in the existing scheme, having misread the cue because the same red marks both the manual high-risk tier and time-critical semi-automatic tasks, recommending "verschiedene [...] Farben"

(CO 7, 26:02). Read together with CO 8's praise for the clean, low-colour baseline, the signal is not contradictory: the palette should stay reserved but make its few colours semantically unambiguous. CO 2 reported weak confirmation feedback after issuing an assignment, as the resulting page looked too similar to the previous one, leaving him unsure whether he had acted. CO 7 also flagged two unprompted data-quality gaps at registration: the single name field invites first-name-only entries (CO 7, 9:03), and the absence of email confirmation leaves the system open to bot accounts (CO 7, 11:59). Localisation defects were also noted: the US date format confused CO 3, and CO 4 identified the English-only interface as an outright adoption blocker for an older, less technical volunteer base, insisting the system must be in German. A recurring request for organisation-configurable task categories was raised by CO 2, CO 4, and CO 7 alike.

### 6.1.3 Trust and the Human-in-the-Loop architecture

Coordinators' trust in the match score consistently derived from retained control over the criteria rather than from the score itself. CO 2 articulated this most precisely: he could trust the system "weil ich es ja mitdefiniere" (CO 2, 46:43), while explicitly noting that this trust did not extend to the correctness of the applicants' self-reported data. CO 7 gave the most explicit statement of the mechanism, explaining that her initial trust was lower because someone else had built the ranking, but "wenn ich selber dahinter stecke und das Ranking mache [...] dann würde ich glaub schon dem [...] Ranking vertrauen" (CO 7, 52:18).

CO 1 likewise rated the score positively conditional on configurable criteria, underwritten by the manual mode as a fallback, and CO 8 noted that score quality "hängt natürlich von der Qualität der Daten davor ab" (CO 8, 42:01). CO 4 endorsed the specific weighting of skills above location on domain grounds, and CO 3 went furthest, suggesting that once the criteria are transparent he "könnte man es wahrscheinlich vollautomatisch durchlaufen lassen" (CO 3, 41:39). This remark, while affirming trust, also challenges the necessity of the semi-automatic middle tier for an experienced operator.

The override function was widely endorsed as a control mechanism, described as "Sehr wertvoll" (CO 1, 32:57) and "Sehr praktisch" (CO 5, 30:14), and was the source of a strong sense of agency for CO 2, who described remaining "Chef des Teams" (CO 2, 48:49). However, the override was also the single most-criticised feature across the set, with the critiques pointing in compatible directions. Several coordinators argued that it is only meaningful when paired with supporting information: removing an unknown applicant without a profile to inspect or an articulable reason degenerates into arbitrary capacity trimming (CO 2, CO 3), and CO 6 added that one-time removal is insufficient without a permanent block-list for unsuitable helpers. CO 8 raised the absence of an undo or waitlist, asking "kann ich die Personen zurückholen, die ich schon aussortiert habe?" (CO 8, 43:35) before accepting that she would have to generate a new request. Most pointedly, CO 4 challenged the very semantics of the override in the volunteer domain. Because "man hat immer zu wenig Freiwillige" (CO 4, 57:29), she "würde nie jemanden rauskicken" (CO 4, 57:56) but would re-route applicants elsewhere, objecting that the removal action means "die sind dann weg" (CO 4, 58:05). This connected to her dominant and most emphatically repeated theme, which also surfaced as the headline missing feature: the absence of any appreciative communication to applicants. CO 4 insisted that every applicant, accepted or rejected, requires a personalised, appreciative message, since silent decline "wirft [...] ein schlechtes Bild auf die Organisation" (CO 4, 32:39) and the entire system depends on volunteers acting in their free time. This converges with a notification gap independently raised by CO 1, CO 2, and CO 3, who all wanted clearer closure of the loop after a decision.

A further trust theme came from CO 8, who articulated most clearly what the human-in-the-loop design is for: the system cannot capture the relational and biographical context a coordinator carries, such as knowing that a volunteer "wäre aber schon bereit, einen weiteren Einsatz zu machen, obwohl sie jetzt sagt, sie kann zweimal pro Woche" (CO 8, 42:41). She framed this not as a flaw but as the reason the coordinator must remain in the loop, endorsing the design's premise while marking the limit of what automation can represent.

### 6.1.4 General feedback and missing features

Beyond the notification gap, the most convergent feature requests concerned the volunteer-side input step. CO 1 wanted organisations to be able to upload onboarding and safeguarding material delivered to volunteers at registration; CO 3 proposed a data-honesty prompt at registration to address his stated biggest risk of inaccurate self-reported skills; CO 4 listed concrete domain prerequisites such as a criminal-record extract, a code of conduct, and insurance; and CO 7 proposed that manual-mode credentials be uploaded with the application rather than chased afterwards, since an unqualified applicant could then be declined without any outreach: "dann müsste ich [die] gar nicht kontaktieren" (CO 7, 31:58). A recurring scoping observation also emerged from the single-task organisations: both CO 5 and CO 7 judged the three-tier architecture more than their own workflow needed, with CO 7 stating that "für die Glückskette reicht das das Level 3" (CO 7, 45:16) and locating the design's value in mixed-risk organisations such as the Red Cross. CO 7 further noted that the current single-recurrence model does not fit a deployment spanning four cities and several daily shifts, forcing redundant task creation (CO 7, 18:05). Two further adoption risks were raised that good usability alone cannot resolve. CO 6 identified per-user pricing as the recurring deal-breaker for unpaid volunteer work, stating "es scheitert am Schluss immer im Pricing" (CO 6, 48:02); CO 4 stressed that "das System muss schon eingeführt sein, bevor die Krise kommt" (CO 4, 58:54); and CO 7 anticipated a transitional support burden from login fatigue among occasional volunteers accustomed to account-free tools such as Doodle (CO 7, 50:21). Recurring requests for shift-based scheduling (CO 2, CO 7) and a photo of the applicant (CO 4, CO 5) were also noted.

CO 8 concluded that "es wäre ein Gewinn für viele Organisationen" (CO 8, 49:33), and CO 7, after mapping the prototype onto her current workflow and finding "mir fehlt eigentlich nichts" (CO 7, 1:00:41), expressed interest in adopting it for Glückskette once it reaches production maturity, one of the strongest stated adoption interests in the set.

## 6.2 Volunteers

The findings below are organised by the themes of the volunteer interview guide (see Section 3.4). Direct quotations are given in the original German; interviewee codes follow the scheme in Table 3.

### 6.2.1 Prior volunteering experience

Three volunteers (V1, V3, V4) described prior systems that were cluttered, slow, or informal. V1 contrasted her old platform of "über 1000 Tabs und Links" (V1, 13:59) and long wait times with the prototype's single-page layout, V3 recounted a placement process that worked only "durch Beziehungen" (V3, 11:24) and was not at all uniform, and V4 described a phone-coordinated COVID test-centre where he had been double-booked, cancelling an important personal appointment for a slot that was no longer needed when he arrived (V4, 20:36).

Time pressure on the volunteer side took two distinct forms. V1 framed it as expected response latency, since for short, casual tasks she "erwartet [...] schon schnell eine Antwort, damit man auch seine Zeit gut einplanen kann" (V1, 14:48). V3 located it in a hard external deadline, describing the placement search as "stressig" (V3, 12:48). V2, by contrast, reported no time pressure in her steady-state Street Church work, with placements running on roughly a month's notice (V2, 14:05). This mirrors the coordinator-side scoping observation that the time-pressure framing depends on the type of volunteering activity.

### 6.2.2 Perceived usability and UI design

The application workflow was experienced as effortless across the set. V1 captured this with "Ah, das geht ganz einfach" (V1, 9:17), praising in particular that everything sat on one page rather than scattered across links and tabs. V3 explicitly endorsed the minimal aesthetic, stating "das gefällt mir minimalistisch" (V3, 10:20), and V4 described the simulation as "super einfach" (V4, 18:39). V2 likewise found the login "extrem einfach" (V2, 19:13), though she immediately flagged a design tension worth recording: the same low-friction signup that pleased her also invites spam or bot accounts, leading her to propose mandatory email confirmation at registration. The same point was raised independently by CO 7 on the coordinator side, making it a robust, however out of scope, cross-interview signal. Three volunteers commented directly on the amount of registration data required. V1 and V2 judged it appropriate and well-justified, with V1 noting that "hier weiss man ja auch, wieso man diese Daten braucht" (V1, 18:03) and V2 stating simply "Datenmenge gut, sehr gut" (V2, 20:01). V3 was the dissenting voice, describing it as "bisschen mehr als üblich" (V3, 14:56) and proposing that birthday in particular should not be mandatory. The disagreement is mild but worth noting.

A small set of concrete UI requests emerged that should be read alongside two early features that the prototype subsequently gained. V1, in the earliest volunteer interview, named two priorities that did not yet exist at the time: a filter to hide tasks she did not want and the ability to withdraw a pending application (V1, 21:35). By the time of V3 and V4, both had been implemented, with V3 explicitly noting the withdraw button (V3, 7:08) and V4 reacting positively to the skills filter with "Ja, cool" (V4, 9:16). Beyond that, the requests were targeted: V2 asked for the location field to link to a map URL and for the detail boxes to be aligned to equal height (V2, 25:42), and V3 requested that the Details and Apply buttons be "50% grösser" (V3, 16:42), echoing the larger-font requests from CO 4 and CO 5.

### 6.2.3 Status feedback and administrative experience

The single most convergent volunteer finding was a gap in status feedback after a coordinator decision. V2 said: "Das sieht man nicht, dass man abgelehnt wurde" (V2, 9:42). V4 framed it as his top priority, asking that "eine Meldung kommt, dass man nicht angenommen wurde und nicht einfach, dass nichts erscheint" (V4, 21:13). V3 surfaced the same underlying issue from a different angle, reporting that he had to manually refresh the browser to discover he had been accepted because the dashboard did not update automatically (V3, 16:10). V1 grounded the same theme in her prior experience, where prolonged silence after an application had made her doubt the legitimacy of the organisation altogether (V1, 20:36). Across all four interviews the message is consistent: the prototype currently lacks proactive status communication. This converges directly with the coordinator-side notification gap raised by CO 1, CO 2, CO 3, and most strongly by CO 4.

A second, narrower trust theme came from V2, who articulated a competency-verification concern from the volunteer's own perspective. She asked how the system could ensure that an applicant claiming to be qualified for sensitive work, in her example psychological support, actually was: "Nicht jeder, der etwas angibt, kann das auch" (V2, 20:33). She proposed CV upload at registration as a self-serve evidence mechanism, since under time pressure coordinators "haben nicht Zeit, alles telefonisch zu kontaktieren und abzuholen" (V2, 29:13). This is the volunteer-side mirror of the data-quality concerns raised by CO 2, CO 3, and CO 7, and the proposed fix converges with CO 7's request that manual-mode credentials be uploaded with the application rather than chased afterwards. V3 added a fourth, more specific administrative-experience observation: the paperwork required to verify volunteer hours in his prior placement was "bürokratisch" (V3, 18:35), with more documents than he had expected, an irritation he would want a platform to simplify.

### 6.2.4 General feedback

Beyond the convergent status-feedback gap, the volunteers' top priorities pointed in compatible directions. V1 named the filter and the withdraw function, both subsequently implemented. V2 named email confirmation and CV upload as integrity features. V3 asked that contact information be surfaced prominently on the dashboard rather than buried in settings (V3, 19:13), aligning with CO 5's request for more readily available direct contact. V4's top priority was the status notification itself. Overall verdicts were positive within the limits of a short guided session: V1 closed with "ich finde ein super System" (V1, 26:26), and V3 described the prototype at close as "schon ganz funktionsfähig" (V3, 19:13).

## 6.3 Cognitive Load

### 6.3.1 Coordinators

This section examines whether the prototype achieved its central aim of easing the cognitive burden of volunteer coordination. Two sources inform the assessment: the coordinators' qualitative reflections during the walkthrough and interview, and the NASA TLX ratings, which each coordinator gave verbally and which were then recorded in the NASA TLX application.

As described in Section 3.4, participants were asked throughout to judge the prototype against the tools and practices they already use, so the evidence below speaks to perceived load relative to existing coordination work rather than to load measured in absolute isolation. Ratings were collected on the 0–20 scale used by the NASA TLX application, which allowed the responses to be exported directly, and were then normalised to the standard 0–100 TLX range for reporting (Cao et al., 2009). Because participants rated the prototype relative to the systems they already use, the scale midpoint of 50 represents parity with that existing baseline, so any score below 50 indicates a perceived improvement over current practice.

Figure 12 reports the six raw dimension subscales for each coordinator, and Figure 13 the overall Raw TLX, the unweighted mean of the six subscales, on the full 0–100 range. The overall scores sit consistently in the lower half of the scale, and the per-subscale view shows the same pattern: ratings concentrate toward the low end, with effort and mental demand carrying most of what load there is, which reflects the genuine evaluative work of judging candidates that the system supports rather than removes. (Performance is reverse-anchored, so a low score there denotes a favourable outcome, consistent with the other subscales.)

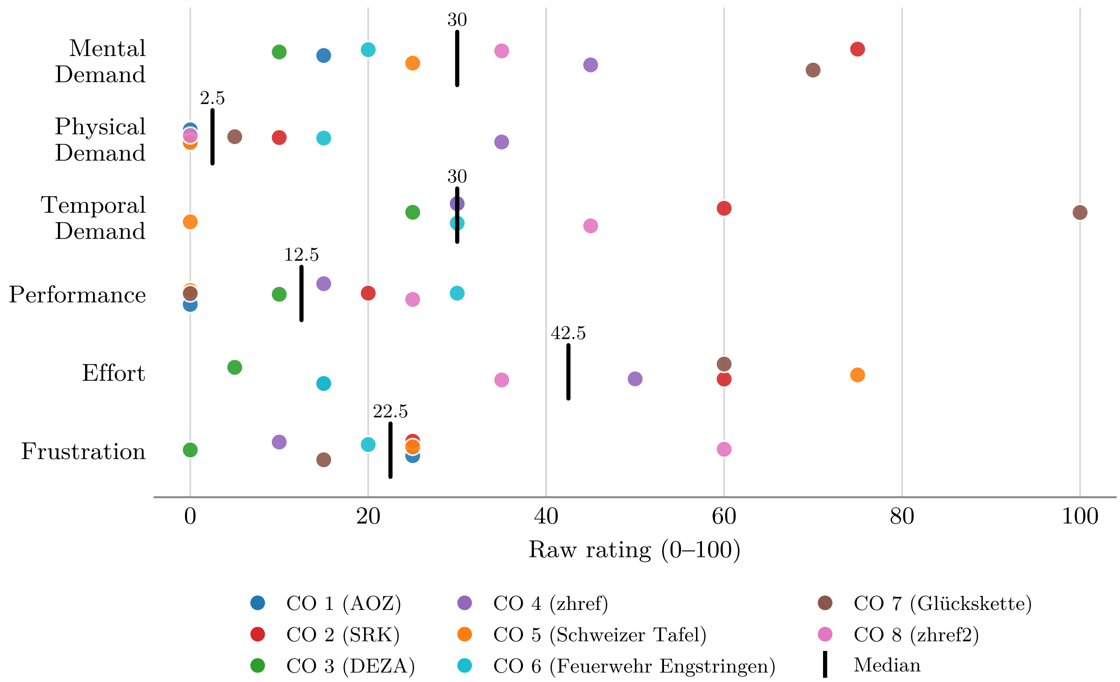


Figure 12: Evaluation across the six NASA TLX categories for coordinators

The semi-automatic ranking was the feature most consistently credited with easing the workload at volume. CO 2 reported that even a two-criterion pre-sort makes the work easier, since the ranking already signals what is relevant, "Das hilft schon sehr" (CO 2, 42:04). CO 4 confirmed that a ranked "Hitliste" makes high-volume handling easier, and CO 3 described the funnel effect concretely, observing that from hundreds of applicants down to the roughly ten needed "kann man schon ziemlich stark reduzieren" (CO 3, 38:12).

The strongest single endorsement came from CO 5, who described the automatic mode as "extrem entlastend" (CO 5, 28:09) set against her current practice of looking up an address and writing an individual email to each applicant. CO 8 captured the same benefit in human-versus-machine terms, describing the system as "eine Hilfe" that handles "vieles sehr schnell [...] und auch fehlerfreier" than she could by hand (CO 8, 40:30), and adding that it is "sicher eine Zeitersparnis" wherever speed matters (CO 8, 41:32). Several coordinators also reported no overwhelming or demanding moments during the walkthrough, with CO 3 summarising the experience as "sehr intuitiv, einfach rein" (CO 3, 40:08).

The comparative framing makes the result sharper than the absolute numbers alone suggest. CO 7 judged the prototype lower in mental demand than her current Doodle-based workflow for the equivalent task (CO 7, 56:41), and CO 8 was the most explicit, rating mental demand far lower once expressed relative to her in-house tool and rating effort equally low "weil dieses System alles für mich macht" (CO 8). Read against the baselines participants actually work with, the prototype was experienced as light, and lighter than the tools it would replace.

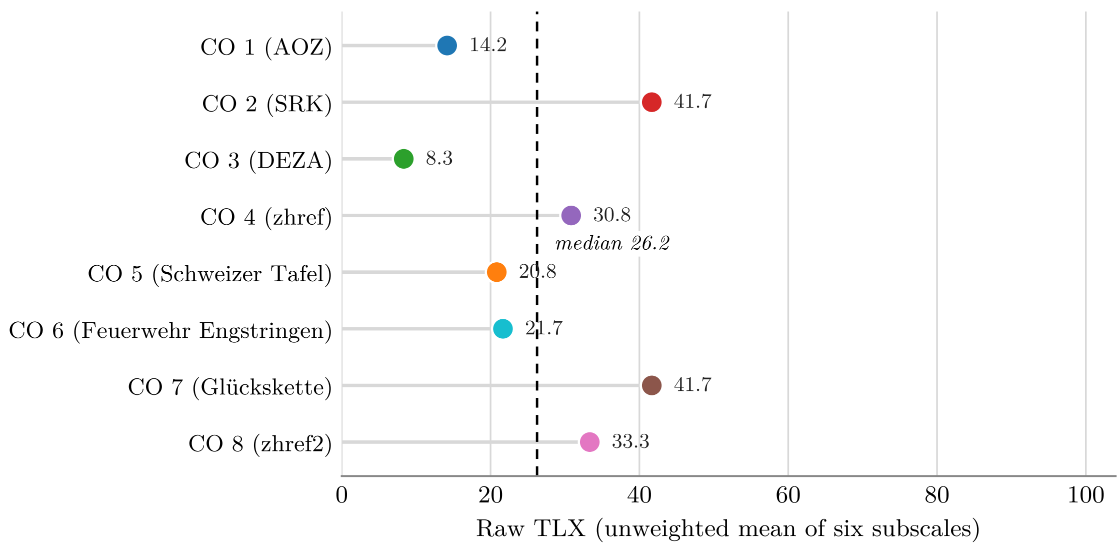


Figure 13: Overall TLX ratings per coordinator

These favourable results come with one honest qualification about their strength rather than their direction: the clearest statement came from CO 6, who held that load reduction was "in so einer kurzen Zeit, mit so wenig Datensätzen, schlichtweg nicht überprüfbar" (CO 6, 42:04), and CO 7 and CO 8 likewise noted that a firm judgement would require working with the system over a longer period and at real operational volume. This bounds what a short, guided walkthrough can demonstrate:

it is well suited to surfacing perceived load and direct comparisons with familiar tools, both of which came out favourably here, but it cannot stand in for a longitudinal measurement of sustained load under field conditions.

Taken together, the consistent qualitative endorsements and the low, comparatively framed TLX scores converge on a clear and encouraging signal, namely that coordinators experienced the prototype as low in perceived load and lighter than their current tools, while pointing to sustained real-world use as the natural next step for confirming it.

### 6.3.2 Volunteers

The four volunteers completed the same NASA TLX instrument under the identical procedure used for the coordinators: ratings were given verbally, recorded in the NASA TLX application, and normalised to the 0–100 range, and participants were again asked to judge the prototype against the methods and systems they had used before, so that the midpoint of 50 represents parity with their prior experience. Because the volunteer interface is deliberately minimal, surfacing only what a volunteer needs to register and apply, the relevant question here is not whether automation reduced their workload but whether the interface imposed any meaningful load in the first place.

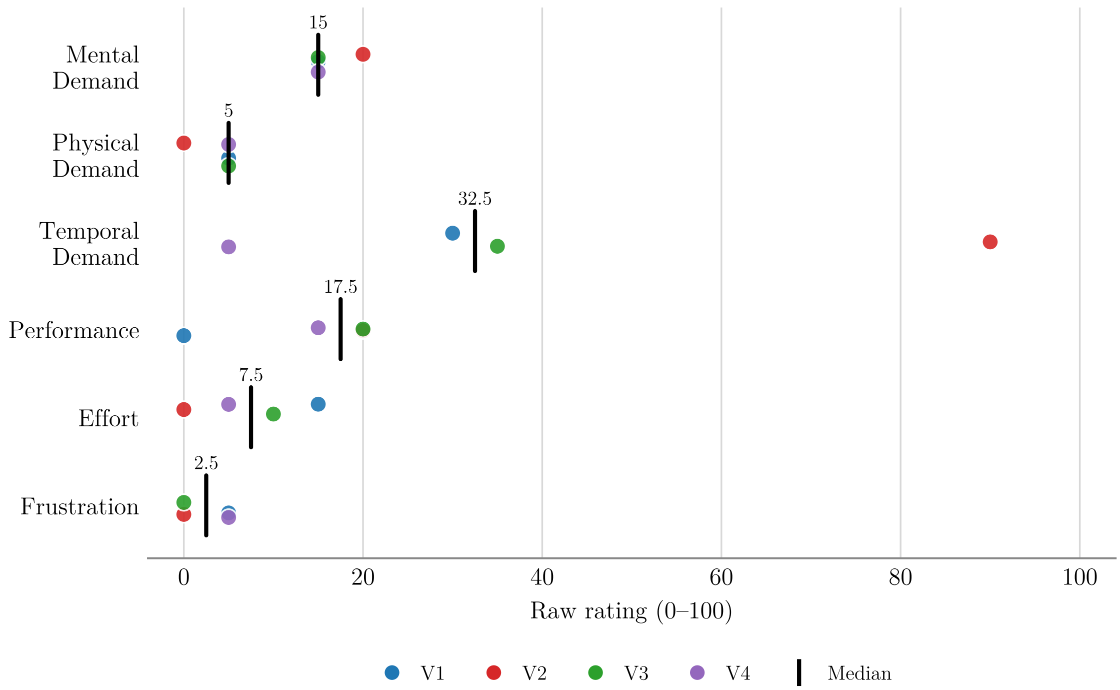


Figure 14: Evaluation across the six NASA TLX categories for volunteers

It did not. Figure 14 reports the six subscales per volunteer and Figure 15 the overall Raw TLX. Perceived load was uniformly low and sat well below the parity midpoint across all four participants, indicating that every volunteer experienced the prototype as lighter than the systems they had used before. The pattern was consistent rather than driven by any single favourable response: mental demand, effort, and frustration clustered close to the floor throughout, and none of the volunteers reported an overwhelming or demanding moment during the walkthrough, with V3 answering the prompt simply, "Nein, nix" (V3, 17:49). This convergence is notable given that the four came to the task from different starting points, from prior crisis volunteering to routine placements, yet arrived at the same low ratings. Read against the cluttered and poorly communicated processes they described from their own experience, the consistently low scores reflect an application flow that volunteers found genuinely effortless. The qualitative comments, gathered alongside the ratings, discussed in

Section 6.2, reinforced the same impression of a quick, single-purpose process that asked little of them.

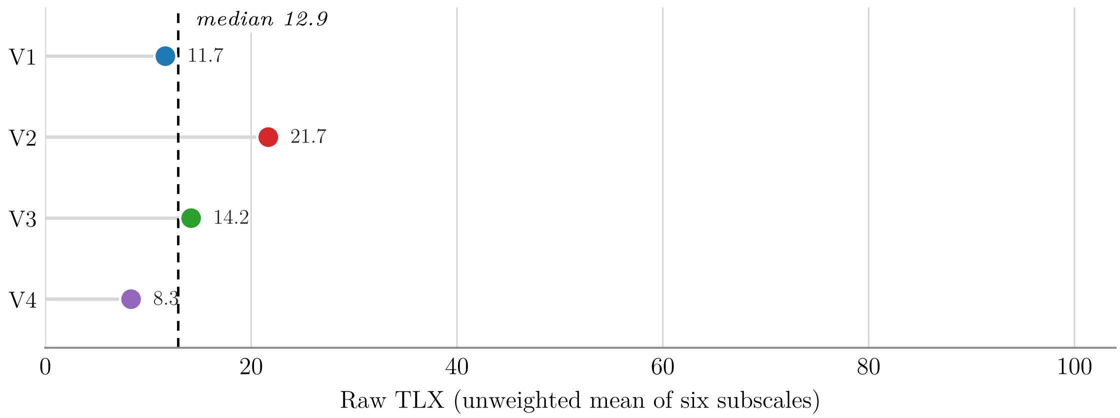


Figure 15: Overall TLX ratings per volunteer

Across the six subscales the same picture holds. Physical demand, effort, and frustration sat at or near the floor for every volunteer, and mental demand was similarly low throughout, so that the overall Raw TLX scores were among the lowest recorded in the study. The only dimension to register any appreciable variation was temporal demand, which accounts for nearly all of the modest spread between participants while the remaining five subscales stayed uniformly low. Within that dimension, V2 produced a single pronounced outlier for which the interview did not surface a specific reason, whereas the other temporal readings remained low.

Even with that variation included, every volunteer's overall score remained well under the parity midpoint, so the aggregate result is unambiguous: the registration and application process was experienced as light, leaving the cognitive design effort of the thesis concentrated where it belongs, on the coordinator side.

## 6.4 Iterative Refinements

The refinements in the prototype were a continuous process, running alongside the interviews, which spanned almost two months. As recurring observations emerged across successive interviews, changes that were both convergent and feasible within scope were integrated back into the prototype before the next sessions, while other suggestions were set aside, either because they exceeded the scope of a thesis prototype or because they came from a single participant without wider corroboration. A main theme was the necessity of recurring tasks. These were introduced by adding a field to the data model, in which the repeatability is set to `none`, `daily`, `weekly`, `every 2 weeks`, `monthly`.

Staying with the task improvements: early in the evaluation there were multiple requests for clearer feedback on task capacity. This was resolved by adding a slider for the capacity overview, with the task turning green once filled to full capacity. Filtering was strengthened on both sides of the system, with a task filter by postal code and by skills and capabilities added for coordinators and volunteers alike. On the coordinator side it sharpens the reduction from a large applicant pool toward the few actually needed, and on the volunteer side it lets applicants surface relevant tasks without scanning the full list. On the volunteer side, a `Show tasks matching my skills` shortcut was added that pre-populates the filter directly from the volunteer's registered profile in a single tap, collapsing the most common filtering intent to minimal input. Volunteers were also given more control over their own applications, in that a pending application can be withdrawn at any time as long as no decision has yet been issued, which respects the volunteer's autonomy and reduces stale entries on the coordinator side. The event log was made more accessible by placing a direct link in the top bar next to the other shortcuts. The main design adjustments concerned the colours and framing of the buttons, improving usability in line with stakeholder wishes.

# 7 Discussion

This chapter returns to the research question posed in Section 1.3 in light of the evidence gathered in Chapter 6. The interpretation that follows weighs the qualitative feedback and cognitive load results against the design's central premises, before stating what the work contributes, where its claims are threatened, what it does not cover, and where it points next.

## 7.1 Interpretation of Results

The results confirm that the prototype lowers perceived cognitive load, but the more useful question is where that effect comes from. The thesis assumed the coordinator's main problem is deciding too fast under pressure. For much of the sample this was not the real bottleneck. Several organisations were rarely overwhelmed by the speed of the decision and far more often by the quality of what fed into it, namely incomplete profiles and unverified applicants. What the dashboard does well is pull scattered information into one clear view, and that benefit holds whether or not a crisis clock is running. Consolidation does not resolve the data-quality problem, but it does relocate it.

Gaps that scattered tools left invisible until deployment become apparent at the point of decision, which is the first condition for addressing them, though the correction itself still depends on better capture upstream. The artifact therefore earns its value less through urgency relief than through consolidation and filtering. Time pressure stays accurate at the crisis and high-volume end, but as a universal motivation it is narrower than first claimed. This was also hard to test directly, since, fortunately, no major crisis was unfolding during the evaluation. The coordinators who had worked in actual crises, however, found the approach a good fit for those conditions, which is the strongest signal available short of a real deployment.

The most important reading is where the Cognitive Load framing runs out. CLT treats the coordinator as a working-memory-bound information processor, and the interface was built around that. The evaluation showed a side the model does not capture, where a real part of the work is relational rather than computational, about keeping volunteers willing and not turning them away. The friction around the override is the clearest example. Removing a weaker candidate makes sense in a selection task, but it is almost incoherent under chronic volunteer scarcity, where the sensible response is to reroute rather than reject. A purely load-oriented account cannot explain this, because the cost being managed is goodwill, not mental effort. The design reduced the load it set out to reduce, but load is only one part of the problem.

These readings sharpen the answer to the research question rather than overturn it. A switchable, per-task level of automation proved to be a sound and well-received basis for the application, and its flexibility was exactly what let it fit very different organisations.

Those with genuinely mixed task risk gained the most from moving between tiers, while even single-profile operations could simply settle on the one mode that suited them. This adaptability is the design's central strength. Improved crisis response, on this evidence, comes less from any fixed level of automation than from giving coordinators the means to set it to their own risk landscape, and the prototype shows that this can be done in a way practitioners find both usable and trustworthy.

## 7.2 Contributions

This thesis makes three contributions to the design of volunteer coordination systems in crisis contexts, answering the research question posed in Section 1.3.

The first is a functional prototype, CrisisMatch, that operationalises a three-tier Human-in-the-Loop architecture for spontaneous volunteer matching. Unlike prior approaches that treat automation as a binary or globally fixed property, the system implements automation as a per-task configuration, letting coordinators move between manual, semi-automatic, and fully automatic assignment modes based on situational demands and task risk. A structured Task Risk Assessment and Automation Mode Evaluator supports this choice by scoring four risk dimensions against task urgency, so the system not only offers a switchable degree of automation but actively guides coordinators toward an appropriate one. This addresses the gap identified in the related work review, namely the absence of a practical, switchable HITL system that balances coordination efficiency with the contextual trust real-world crisis response requires.

The second contribution is empirical. Across twelve semi-structured interviews (eight with coordinators from seven organisations, four with volunteers), coordinators experienced the consolidated dashboard, the progressive disclosure structure, and the automation wizard as less mentally demanding than the tools they currently use, see Section 6.

The interviews also added a practical dimension to the theoretical account that motivated the design. Beyond this, they revealed that a strong driver of acceptance was authorship of the matching criteria. Coordinators accepted the automated ranking less because of the results it produced and more because they knew that, in a real deployment, they would be able to define those criteria themselves. This was clearest in the semi-automatic mode, where a ranked shortlist they could overrule was treated not as something imposed but as an extension of their own judgement once the prospect of control was on the table.

A second observation is that the system eased perceived load even with no time pressure present, because consolidating everything into one view surfaces missing or incomplete data at the moment of deciding rather than only later during deployment. Together these results shift how the three-tier design should be understood. Its real value is adaptability, fitting organisations with very different needs through a single system, rather than speed in a crisis.

The third contribution is conceptual. The evaluation surfaced a limit of the cognitive-load framing that motivated the design. Coordinating unpaid volunteers is not only an information-processing task but a relational one, in which preserving volunteer goodwill can matter as much as reducing mental effort. The friction around the override function, logical as a selection mechanism yet difficult to justify under conditions of chronic volunteer scarcity, shows concretely where a load-only account fails to fully capture the nature of the work. Identifying this relational dimension is a contribution that holds independently of the prototype and indicates how future tools in this space should be framed.

## 7.3 Threats to Validity

**Internal Validity:** The evaluation relied on structured walkthroughs rather than real crisis deployments. Participants interacted with a prototype under low-pressure conditions, which may have led them to assess the system more favourably than they would under the acute time pressure and incomplete information that characterise actual disaster response. Perceived cognitive load may therefore be underestimated relative to real-world use.

**External Validity:** The coordinator sample consisted of eight coordinators from seven organisations, several of which had not been directly involved in acute crisis coordination. Their evaluation of the prototype was therefore partly based on projected rather than lived operational experience, which may have led to more favourable usability assessments than practitioners with direct crisis exposure would produce.

**Construct Validity:** A related threat lies in the NASA-TLX instrument as it was applied here. Participants were asked to rate the prototype relative to the systems they already use, which makes the baseline subjective and specific to each individual rather than a fixed reference point. The reading of the scale midpoint as parity with existing practice is therefore an interpretation of the comparative framing rather than a calibrated measurement, and the absolute values should not be compared across participants as if they shared a common zero.

**Evaluator Bias:** All interviews were conducted by the researcher who also designed and built the prototype. Participants may have moderated critical feedback as a result, particularly given the collaborative framing of the sessions.

## 7.4 Limitations

**Matching algorithm:** The semi-automatic and automatic modes rank candidates based on skill match and location proximity, implementing a multi-objective filter within the scope of this thesis. However, the weighting and prioritisation of these parameters was defined by the researcher rather than being configurable by the coordinator. How coordinators would define and adjust these criteria in practice, and whether the default weighting holds across different crisis types, is left open and addressed further in Section 7.5.

**Single-session evaluation:** Each participant interacted with the prototype once. Longitudinal effects, such as skill degradation in manual mode after extended use of the automatic mode, a concern raised explicitly by Parasuraman et al. (2000) and Onnasch et al. (2014), could not be observed. The evaluation therefore captures first impressions rather than the behavioural consequences of sustained use.

**Scale:** The volunteer-facing interface was evaluated with four participants, all of whom responded positively overall. While this provides initial confidence in the design, the prototype has not been tested at scale. The core matching logic is not expected to change, but performance, data quality, and edge cases under a large influx of simultaneous volunteers remain untested.

**Data authenticity:** The matching logic operates on the profile data volunteers provide about themselves, and the system takes this input at face value. Skills, qualifications, and availability were assumed to be accurate, with no verification mechanism in place. In a real deployment the reliability of a match depends directly on the truthfulness of this self-reported data, and on volunteers following through on their commitment, neither of which the current design checks.

## 7.5 Future Work

The most immediate extension is making the matching criteria configurable by the coordinator. This includes not only the weighting of parameters such as skill match and location proximity, but also the underlying categories themselves: which skills are available for volunteers to declare, what task types and requirement fields coordinators can define, and how these are structured per organisation. Different organisations operate with fundamentally different role vocabularies, and a fixed schema limits how well the system can adapt to contexts beyond the one it was designed for.

A second direction concerns scaling the system for real deployment. Future work should focus on optimising the system architecture and deployment infrastructure to handle large-scale data reliably, including how task and volunteer data is ingested, validated, and kept consistent under high-load conditions. Volunteer verification and the relational dimension of coordination also remain open. The system currently takes self-reported skills and availability at face value, and future iterations could explore lightweight verification mechanisms, such as post-assignment feedback loops or credential uploads, without creating barriers that deter spontaneous volunteers from registering. Beyond verification, the evaluation surfaced that coordination is not only an information-processing task but a relational one, and future tools should explore how design can support volunteer goodwill and retention alongside operational efficiency.

A further direction concerns interaction design within the automation tiers, a dimension distinct from the degree of automation that this thesis varied. Within the human-led tiers, candidate information can be presented in different ways, and recent experimental work suggests this presentation can itself shape how actively the operator engages: Maier et al. (2025) found that prompting users with reflective questions sustained engagement and perceived ownership more than presenting finished suggestions, despite comparable system capability. Whether an inquiry-driven presentation of candidate matches would further reinforce the coordinator engagement that this thesis found central is a question this work leaves open.

# 8 Conclusion

How much of a decision should a machine make? For volunteer–task matching in a crisis, this thesis argues the honest answer is that it depends, and that the choice belongs to the coordinator rather than the system designer. The obvious move, faced with the pressure to match people to tasks quickly, is to automate the bottleneck away and let an algorithm sort the crowd. The more interesting answer turned out to be more human-centred, like most realities in today's world. What coordinators needed was not a system that decides for them, but one that lets them decide how much it decides.

That is the idea CrisisMatch puts to the test, by making the degree of automation a property of each task rather than the whole system. The instinct to keep the human in control held up, but the evaluation rearranged the reasons it works. The thesis began from time pressure, yet that was rarely the bottleneck practitioners described. What carried the design was steadier than urgency. A single consolidated view replaced scattered tools, so that missing or unreliable information surfaced at the moment of deciding rather than later in the field, and the prospect of owning the matching logic was what turned a ranked shortlist from something imposed into an extension of the coordinator's own judgement. The same dial that let one organisation hand-pick a specialist let another batch-fill a hundred shifts, and that adaptability is what fit a single system to organisations with very different needs.

The work is honest about where it stops. It showed that coordinating volunteers is as much about goodwill as about logistics. A discarded helper is not a closed ticket but a person less likely to return, and no amount of matching repairs a profile that was never true to begin with. These are not cracks in the approach so much as the terrain it has to be built for. Within that terrain the thesis makes a modest but defensible claim. The right amount of automation for crisis volunteering is not a number to be found once and fixed, but a dial worth handing to the person holding the crisis, and a system that trusts them to turn it is one they are willing to trust back.

# 9 Usage of Artificial Intelligence

Artificial intelligence tools were used at several stages of this thesis, always under the direction and final judgement of the researcher. Their use per phase is summarised in Table 4.

| Phase | Use and Researcher Oversight |
|---|---|
| Literature research | AI models, such as Perplexity Sonar, helped identify and surface potentially relevant academic papers. All sources were read, assessed, and selected manually. |
| Prototype implementation | Agentic AI, primarily Claude Opus, generated boilerplate code, assisted with debugging, and implemented much of the interface's visual language. The design intent, including layout, interaction patterns, and the colour vocabulary (Section 5.4.6), was specified by the researcher. The AI executed these decisions rather than originating them. |
| Internal literature tool | A custom system tokenized and vectorized the collected papers for semantic search across the literature, used primarily to verify and locate existing claims rather than to generate text. Interview data was deliberately excluded to protect participant confidentiality. |
| Writing | AI assistance was limited to correcting grammar and, in some passages, refining stylistic phrasing, for example using Claude Sonnet. All ideas and arguments originate with the researcher and were drafted manually. |

Table 4: Usage of artificial intelligence by phase

No part of the thesis's intellectual content was produced by AI.

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# A Appendix 1

## A.1 Complete Requirements

### A.1.1 Functional Requirements

Table 5: Functional Requirements

| ID | Requirement | Source | Priority |
|---|---|---|---|
| FR-01 | The system shall provide a virtual VRC as a web-based platform that enables volunteers to register. | (FEMA, 2014) | Must |
| FR-02 | The system shall display current volunteer opportunities and tasks related to the incident. | (FEMA, 2014) | Must |
| FR-03 | The system shall support rapid registration by collecting only essential volunteer data needed for matching and operations. | (FEMA, 2014) | Must |
| FR-04 | The system shall allow volunteers to enter their own registration data via self-assessment. | (Betke et al., 2023). | Must |

*Continued on next page*

*Table 5 continued*

| ID | Requirement | Source | Priority |
|---|---|---|---|
| FR-05 | The system shall allow coordinators to add volunteers who cannot register themselves. | (Betke et al., 2023; Rauchecker & Schryen, 2018) | Should |
| FR-06 | The system shall allow coordinators to create tasks with schedule, priority, description, activities, per-activity demand, location, and optional meeting point. | (Betke, 2018; Rauchecker & Schryen, 2018) | Must |
| FR-07 | The system shall allow coordinators to edit task information as the situation changes. | (Rauchecker & Schryen, 2018) | Should |
| FR-08 | The system shall implement an algorithmic method to generate volunteer–task assignment proposals. | (Petrovic, 2026) | Must |
| FR-08a | The baseline matching method shall be bipartite matching or weighted scoring. | (Petrovic, 2026) | Could |

*Continued on next page*

*Table 5 continued*

| ID | Requirement | Source | Priority |
|---|---|---|---|
| FR-09 | The system shall support manual, semi-automatic, and automatic assignment proposal modes. | (Betke et al., 2023; Petrovic, 2026) | Must |
| FR-10 | In semi-automatic mode, the system shall present coordinator-reviewable proposals before notifying volunteers. | (Betke et al., 2023) | Should |
| FR-11 | Regardless of the automation mode, volunteer agency shall be preserved throughout the assignment process. The system shall require volunteers to apply for open tasks themselves, ensuring that no assignment is confirmed without an initial act of self-selection on the volunteer's part. | (Betke et al., 2023) | Must |
| FR-12 | The system shall restrict assignments to volunteers who match the required skill constraints for a task. | (Rauchecker & Schryen, 2018) | Should |

*Continued on next page*

*Table 5 continued*

| ID | Requirement | Source | Priority |
|---|---|---|---|
| FR-13 | The system shall restrict assignments to time blocks where volunteers are available. | (Rauchecker & Schryen, 2018; Falasca et al., 2009) | Should |
| FR-14 | The system shall prevent assigning a volunteer to multiple concurrent task activities. | (Rauchecker & Schryen, 2018) | Should |
| FR-15 | The system shall allow coordinators to set and enforce a maximum number of shifts per volunteer within a planning horizon. | (Falasca et al., 2009; Rauchecker & Schryen, 2018) | Could |
| FR-16 | The system shall support balancing unmet demand across tasks to avoid critical gaps. | (Falasca et al., 2009) | Should |
| FR-17 | The system shall support distributing volunteers across active tasks to reduce overrun at some sites while others remain in urgent need. | (Betke, 2018) | Should |

*Continued on next page*

*Table 5 continued*

| ID | Requirement | Source | Priority |
|---|---|---|---|
| FR-18 | The system shall allow volunteers to manually update their status (e.g., *Arrived*, *Job Done*) via the web UI. | (Auferbauer et al., 2016; FEMA, 2014) | Won't |
| FR-19 | The volunteer UI shall display only assignment-specific mission details and avoid presenting broader disaster information. | (Betke, 2018; Nielsen, 1993) | Should |
| FR-20 | The system shall allow volunteers to view their current assignment details at any time. | (Betke, 2018) | Should |
| FR-21 | After submitting availability and skills, the system shall acknowledge receipt to the volunteer. | (Betke, 2018) | Should |
| FR-22 | The system shall allow coordinators to send task alerts to all matched volunteers or selected groups simultaneously. | (Betke et al., 2023) | Won't |

*Continued on next page*

*Table 5 continued*

| ID | Requirement | Source | Priority |
|---|---|---|---|
| FR-23 | The system shall provide a back-channel mechanism to communicate with registered volunteers (e.g., email or web UI notifications). | (Windsperger, 2025; FEMA, 2014) | Could |
| FR-24 | The system shall support documenting referrals, volunteer activities, and hours worked for reporting and reimbursement. | (FEMA, 2014) | Could |
| FR-25 | The system shall support issuing and recording a digital form of volunteer identification approved by local officials. | (FEMA, 2014) | Won't |
| FR-26 | The system shall allow coordinators to flag screening requirements per task or volunteer and record screening status, without implementing the screening process itself. | (FEMA, 2014) | Should |

*Continued on next page*

*Table 5 continued*

| ID | Requirement | Source | Priority |
|---|---|---|---|
| FR-27 | The system shall support coordinator search and filtering of volunteers by skills and availability to support manual matching. | (Windsperger, 2025; Petrovic, 2026) | Could |
| FR-28 | The system shall allow coordinators to set volunteer capacity limits per shift to automatically distribute workload. | (V_Helper_I01: 00:07:58) | Must |
| FR-29 | The system shall allow coordinators to copy shift structures to future days, reducing repetitive manual schedule creation. | (V_Helper_I01: 00:32:33) | Should |
| FR-30 | The system shall automatically mark tasks as filled once capacity is reached, preventing double-booking. | (V_Helper_I02: 00:33:12) | Must |
| FR-31 | The system shall provide volunteers with a filtering engine to sort available tasks by category (e.g., location, required skills). | (V_Helper_I02: 00:09:39) | Must |

### A.1.2 Non-functional Requirements

Table 6: Non-Functional Requirements

| ID | Requirement | Source | Priority |
|---|---|---|---|
| NFR-01 | The system shall provide an interface that minimises co-ordinator cognitive load by clearly visualising the real-time status of all tasks. The coordinator must be able to assess volunteer capacity and task shortages at a glance, without cross-referencing external lists. | (Windsperger, 2025) | Must |
| NFR-02 | The system shall remain usable when large numbers of volunteers register in a short period. | (Betke et al., 2023; Petrovic, 2026) | Should |
| NFR-03 | The system shall make up-to-date task and volunteer data quickly available to support response decision-making. | (Ristvej & Zagorecki, 2011) | Should |
| NFR-04 | The system shall provide traceability via event history to reproduce actions after the incident. | (Betke, 2018) | Could |

*Continued on next page*

*Table 6 continued*

| ID | Requirement | Source | Priority |
| --- | --- | --- | --- |
| NFR-05 | The system shall be able to store and archive data and produce reports based on assignments and hours. | (Ristvej & Zagorecki, 2011; FEMA, 2014) | Could |
| NFR-06 | The system shall provide secure but easy and reliable access control appropriate for crisis operations. | (Ristvej & Zagorecki, 2011) | Should |
| NFR-07 | The system shall implement a fixed, minimal data schema collecting name, email, skills, and location, in adherence to data minimisation principles. | (Betke, 2018) | Must |
| NFR-08 | The web UI shall be responsive for both coordinator and volunteer use across devices. | (Windsperger, 2025) | Must |
| NFR-09 | The system shall be designed so that data maintenance is operationally feasible for the municipality. | (Ristvej & Zagorecki, 2011) | Won't |

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*Table 6 continued*

| ID | Requirement | Source | Priority |
|---|---|---|---|
| NFR-10 | When presenting automatic or semi-automatic matches, the system shall provide a brief explanation of why a volunteer was matched (e.g., matching skills or availability). | (Windsperger, 2025) | Could |
| NFR-11 | The system shall keep interfaces open for possible integration into existing mission command systems. | (Betke et al., 2023; Betke, 2018) | Won't |
| NFR-12 | The system shall support limiting volunteer-facing information to reduce risks from self-coordinated behaviour. | (Betke, 2018) | Should |
| NFR-13 | The prototype shall be structured into three major components: Registration, Matching/Allocation, and Coordinator Dashboard. | (Betke, 2018) | Should |

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*Table 6 continued*

| ID | Requirement | Source | Priority |
|---|---|---|---|
| NFR-14 | The system shall implement a user vetting process to shift the security burden from the coordinator to the system, preventing malicious actors during an acute crisis phase. | (V_Helper_I01: 00:32:35) | Could |
| NFR-15 | The system shall serve as a centralised, verified information repository to reduce the time coordinators spend untangling broken communication chains and correcting rumours. | (V_Helper_I01: 00:20:00) | Must |

### A.1.3 Information Requirements

Table 7: Information Requirements

| ID | Requirement | Source | Priority |
|---|---|---|---|
| IR-01 | The system shall store volunteer identity and contact data (at minimum: name and email address) to enable registration and communication. | (FEMA, 2014) | Must |
| IR-02 | The system shall store volunteers' self-reported skills and capabilities for skill-based matching. | (FEMA, 2014; Petrovic, 2026) | Must |
| IR-03 | The system shall capture a fixed set of capability checkboxes including physical work difficulty levels and information-related work (e.g., hard/medium/easy physical work; paperwork; care; information retrieval). | (Betke, 2018) | Should |
| IR-04 | The system shall capture volunteer availability as time blocks or comparable structured availability entries. | (Falasca et al., 2009; Betke, 2018) | Must |

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*Table 7 continued*

| ID | Requirement | Source | Priority |
|---|---|---|---|
| IR-05 | The system shall capture volunteer task preferences to support preference-aware matching. | (Falasca et al., 2009) | Should |
| IR-06 | The system shall store volunteer location as text-based regions (postal code or distinct identifiers) to support location-aware coordination. | (Betke, 2018) | Could |
| IR-07 | The system shall store volunteer assignment status per task (e.g., proposed, accepted, declined, completed, cancelled). | (Betke, 2018) | Must |
| IR-08 | The system shall store, per task, category, description, location, required skills, and volunteer capacity. | (Windsperger, 2025) | Must |
| IR-09 | The system shall store task start/end time and scheduling windows for time-based matching. | (Betke, 2018) | Must |

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*Table 7 continued*

| ID | Requirement | Source | Priority |
|---|---|---|---|
| IR-10 | The system shall store a task priority level to support prioritised allocation in overload situations. | (Rauchecker & Schryen, 2018; Betke, 2018) | Should |
| IR-11 | The system shall allow tasks to include multiple activities with a per-activity volunteer demand. | (Betke, 2018) | Won't |
| IR-12 | The system shall store volunteer work documentation including hours served and types of work performed. | (FEMA, 2014) | Could |
| IR-13 | The system shall store referral details sent to volunteers, including duties description, full address, and a contact point. | (FEMA, 2014) | Should |
| IR-14 | The system shall store an event history (who did what, when) for later reconstruction and accountability. | (Betke, 2018) | Could |

### A.1.4 Design Requirements

Table 8: HCI Design Requirements

| ID | Requirement | Source |
|---|---|---|
| *1. Limit Working Memory Demands* | | |
| DR-01 | Do not present more than 3–4 interacting information elements simultaneously. Working memory can process only about 3–4 novel elements at a time before capacity is exceeded. Volunteer profiles, tasks, location, and urgency should never all appear as raw data fields simultaneously without visual grouping. | (Sweller, 2011) |
| DR-02 | Chunk related information into meaningful groups (e.g., a Volunteer Profile Card). Users process pre-organised schemas as single units, dramatically reducing working memory load. Group volunteer name, skill tags, availability, and distance into one card component rather than dispersed fields. | (Sweller, 2011) |
| *2. Eliminate Extraneous Cognitive Load* | | |

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*Table 8 continued*

| ID | Requirement | Source |
|---|---|---|
| DR-03 | Spatially integrate related information; never force the user to look in two places at once (Split-Attention Principle). When multiple sources of information must be mentally integrated, they should be physically co-located. Separating them forces unnecessary working memory use. Example: show a volunteer's skills directly on the task they are matched to, not on a separate panel. | (Hollender et al., 2010; Sweller, 2011) |
| DR-04 | Do not repeat the same information in multiple formats simultaneously (Redundancy Principle). Displaying the same data as both a badge icon and a text label when one is self-explanatory forces unnecessary processing and increases extraneous load. Use icons with tooltips, not permanent dual-format display. | (Hollender et al., 2010; Sweller, 2011) |

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*Table 8 continued*

| ID | Requirement | Source |
|---|---|---|
| DR-05 | Minimise extraneous interface features that distract from the core task. Interfaces with unnecessary features cause greater cognitive load, slower task times, and impaired meta-cognition compared to minimal interfaces. Remove decorative elements, redundant labels, and unrelated navigation from the matching screen. | (Oviatt, 2006) |
| DR-06 | Eliminate interruptions and non-essential notifications during active matching tasks. Interruptions directly undermine high-level planning and integrative thinking. The matching workflow should suppress badge pop-ups, background alerts, and unrelated status messages during active use. | (Oviatt, 2006) |
| *3. Reduce Decision Complexity (Hick–Hyman Law)* | | |

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*Table 8 continued*

| ID | Requirement | Source |
|---|---|---|
| DR-07 | Limit the number of presented options per decision step; use hierarchical structuring over flat long lists. Reaction time increases logarithmically with the number of stimulus–response alternatives (Hick's Law). Broader, shallower menu trees yield faster decisions than narrower, deeper ones. Show 5–9 best-matched volunteers per task by default, not all candidates. | (Proctor & Schneider, 2018) |
| DR-08 | Use progressive disclosure: show overview first, reveal details on demand. Progressive classification (overview → filter → detail) reduces the number of active decision alternatives at any moment. Implement a collapsed match card that expands on click, not a full-page profile immediately. | (Nielsen, 2006; Proctor & Schneider, 2018) |

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*Table 8 continued*

| ID | Requirement | Source |
| --- | --- | --- |
| DR-09 | Pre-sort and pre-filter match candidates algorithmically before presenting them to the user. Where set-size effects are inevitable, reducing the visible set via preprocessing cuts decision time without limiting user agency. Present a "Top 5 matches" default, with a "Show more" option. | (Proctor & Schneider, 2018) |
| *4. Support Users' Existing Mental Models* | | |
| DR-10 | Design workflows that match users' pre-existing, familiar interaction patterns. Interfaces that depart from users' existing work practice substantially increase cognitive load and reduce performance, particularly for lower-skilled users. Coordinators who already use chat-style lists or spreadsheets should encounter analogous patterns in the application. | (Oviatt, 2006) |

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*Table 8 continued*

| ID | Requirement | Source |
|---|---|---|
| DR-11 | Reduce memory load by externalising state: users should not need to remember information from one screen to another. A core usability goal is recognition over recall. Example: show the task description persistently while browsing candidate volunteers, rather than requiring the user to return to a previous screen. | (Hollender et al., 2010) |
| *5. Leverage Multimodal Presentation* | | |
| DR-12 | Where possible, distribute information across visual and auditory channels rather than relying on one channel alone (Modality Principle). Presenting verbal and visual information in complementary modalities expands effective working memory capacity by using both the phonological loop and visuospatial sketchpad. Example: pair a map view with an audio-cue alert for urgent task notifications. | (Oviatt, 2006; Hollender et al., 2010) |

*Continued on next page*

*Table 8 continued*

| ID | Requirement | Source |
|---|---|---|
| DR-13 | Avoid presenting lengthy, complex textual instructions in audio-only format. The modality effect reverses when auditory content is long and complex, because it must be held entirely in working memory without the ability to re-scan. Long onboarding instructions or help text must be written, not read aloud. | (Sweller, 2011) |
| *6. Adapt to User Expertise* | | |
| DR-14 | Provide richer guided assistance to novice users; reduce it as expertise grows (Expertise Reversal / Guidance Fading). Instructional scaffolding that is essential for novices becomes redundant and cognitively costly for experts. Implement a first-run onboarding wizard that is dismissible, and allow power users to set a compact view. | (Hollender et al., 2010; Sweller, 2011) |

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*Table 8 continued*

| ID | Requirement | Source |
|---|---|---|
| DR-15 | Provide concrete worked examples during onboarding (e.g., a sample match with rationale). The worked example effect shows that presenting a solution reduces the extraneous cognitive load of searching for one from scratch, accelerating schema formation. A pre-populated example task with sample volunteer matches during onboarding directly applies this principle. | (Sweller, 2011) |
| *7. Support Representational Needs* | | |
| DR-16 | Include the representational formats users need to complete their task (maps, skill tags, calendar). Performance is enhanced when interfaces support the representational systems users rely on: linguistic, diagrammatic, symbolic, rather than forcing translation between formats. The matching interface should include spatial (map), temporal (calendar), and categorical (skill badges) representations simultaneously. | (Oviatt, 2006) |

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*Table 8 continued*

| ID | Requirement | Source |
|---|---|---|
| *8. Usability as a Load-Reducing Mechanism* | | |
| DR-17 | Design for Nielsen's five usability goals: learnability, memorability, efficiency, low error rate, and satisfaction. Traditional usability principles directly reduce the software-induced component of extraneous cognitive load. Each goal maps directly: learnability reduces intrinsic load during onboarding; memorability reduces load for infrequent coordinators. | (Hollender et al., 2010) |
| DR-18 | Allow selection from pre-defined options (dropdowns, tags) rather than free-text entry wherever possible. Menu-based selection eliminates spelling errors and reduces the cognitive cost of formulating input. Skill matching, availability, and location should use structured inputs, not open fields. | (Hollender et al., 2010) |

## A.2 Interview Guide Coordinator

**Bachelorarbeit:** “Volunteer Matching System”
**Datum:** 01.04.2026
**Institution:** Universität Zürich (UZH)

## Checkliste: Vor dem Interview

- Prototyp bereit?
- Raum für Studie bereit?
- Einverständniserklärung zweimal ausgedruckt und Stift verfügbar?
- Gerät für Aufnahme bereit / Teams-Aufnahme bereit?
- NASA-TLX (Papier oder online) bereit?
- Smartphone stummgeschaltet und auf Flugmodus?

## Einführung

- **Persönliche Vorstellung:** Nikola Petrovic, Major Softwaresysteme, Minor Neuroinformatik, schreibe meine Abschlussarbeit.
- **Einverständniserklärung:** Ich möchte das Interview für die spätere Analyse aufzeichnen. Alle Daten werden selbstverständlich vertraulich behandelt, anonymisiert und nur für wissenschaftliche Zwecke in diesem Projekt verwendet.
- Interview wird auf Hochdeutsch stattfinden, ist das ok?
- Ist es für Sie ok, wenn ich die Sitzung aufzeichne?

*Audioaufnahme einschalten / Auf Hochdeutsch wechseln*

- Geschätzte Zeit insgesamt: 30–50 min.
- **Sehr wichtig:** Keine falschen Antworten, alle Ihre Erkenntnisse sind wichtig für unsere Arbeit und werden uns helfen.
- Also selbst wenn Sie denken, dass unsere Lösung nicht gut, nützlich oder ähnlich ist, nehmen wir das nicht persönlich.
- Wenn Sie nicht antworten wollen, ist das natürlich auch ok.

## Einführung in das Projekt

- In vielen Krisen bieten viele Menschen sehr schnell Hilfe an, und Koordinatoren müssen Freiwilligen unter Zeitdruck Aufgaben zuweisen.
- Bestehende Tools sind dafür oft nicht ausgelegt und können viel mentale Arbeitsbelastung erzeugen, entweder ganz automatisch oder ganz manuell.
- In meinem Projekt entwickle ich einen Prototyp einer webbasierten Koordinationsplattform namens *Crisis Match*.
- Es konzentriert sich darauf, Koordinatoren bei der Verwaltung von Aufgaben und Freiwilligen zu helfen und die kognitive Belastung bei der Entscheidung zwischen verschiedenen Zuweisungsmodi (manuell, halbautomatisch, automatisch) zu reduzieren.

## Einführung in den Prototyp

- Es gibt im Wesentlichen zwei Ansichten, auf die Sie in der Anwendung zugreifen können:
  - Als **Freiwilliger** können Sie sich registrieren und für Aufgaben bewerben.
  - Als **Koordinator** können Sie Aufgaben erstellen und verwalten.
- In diesem Modus können Sie die verschiedenen Automatisierungsmodi nutzen.

*Nun möchte ich, dass Sie diese Dinge ausprobieren:*

- Sie übernehmen nun die Rolle eines Freiwilligen / Koordinators in einer fiktiven Krise.
- Ich werde mit Ihnen einen Durchlauf machen, indem Sie mehrere Aspekte des Prototyps durchgehen, wie man es in einer echten Situation machen würde:
  1. Einloggen
  2. Aufgabe erstellen
  3. Eine Ausschreibung von jedem Modus verwalten
- Ausserdem möchte ich Sie bitten, direkt laut auszusprechen, was Sie tun und was Sie denken.
- Also, zum Beispiel: "Ok, ich sehe diese Seite, ich denke, ich muss auf diesen Button klicken, damit ich auf diese Seite komme, aber das verwirrt mich ein bisschen, etc."
- Wir gehen es gemeinsam durch.
- **Und wichtig:** Wir testen hier die App, nicht den Nutzer.

- Also, wenn etwas nicht klar ist, ist das ein Problem der App und ich möchte, dass Sie es mir sagen.
- Durchklicken des Prozesses der Erstellung und Verwaltung von Aufgaben.

## Interview-Fragebogen

### Aktuelles Organisationssystem

- Wie handhabt Ihre Organisation aktuell das Matching und die Koordination von Freiwilligen (während einer Krise)?
- Was sind die Hauptherausforderungen bei der Koordination von Freiwilligen in Ihrer Organisation?
- Wie, wenn überhaupt, beeinflusst Zeitdruck diesen Prozess?
- Wo liegen Ihrer Erfahrung nach die grössten Engpässe oder Herausforderungen in diesem aktuellen Aufbau?

### Wahrgenommene Benutzerfreundlichkeit & UI-Design

- Wie haben Sie den Arbeitsablauf über die drei Aufgaben hinweg erlebt?
- War Ihnen klar, warum sich das System bei der *(Manuellen Aufgabe)* anders verhielt als bei der *(Automatischen Aufgabe)*?
- Gab es Momente, in denen Sie unsicher waren, was als Nächstes zu tun ist? Sagen Sie mir, wo das passiert ist.
- Was würden Sie ändern / anpassen (UI-Design)?

### Kognitive Belastung & Entscheidungsermüdung

- Denken Sie an die traditionelle Art, 50 Bewerbungen manuell zu prüfen. Wie hat das Ranking des Systems mit dem Semi-Automatischen (Modus 2) oder das Akzeptieren im Automatischen (Modus 3) bei den Aufgaben mit hohem Volumen Ihren mentalen Aufwand im Vergleich zur manuellen Erledigung beeinflusst? Wurde es leichter / schwerer, etc.?
- Gab es Momente während der Simulation, die sich überwältigend oder anspruchsvoll anfühlten? Erzählen Sie mir davon.
- Denken Sie an hohe Bewerbungszahlen von Freiwilligen: Wie schneidet diese Herangehensweise, verglichen mit den bestehenden Tools, ab?

### Vertrauen & Human-in-the-Loop (Strukturelles / Psychologisches Empowerment)

- In Modus 2 haben Sie die Kontrolle mit dem System geteilt (es hat gerankt, Sie haben auf Akzeptieren geklickt). Wie haben Sie den Match-Score wahrgenommen?
- Welche Rolle, wenn überhaupt, spielte die Override-Funktion dafür, wie sehr Sie sich im automatischen Modus in Kontrolle fühlten?

### Allgemeines Feedback

- Wenn dieses System morgen in einer echten Krise eingesetzt würde, was ist das grösste Risiko oder die fehlende Funktion, die Sie sehen? Und was ist Ihre oberste Priorität zur Verbesserung dieses Prototyps in der nächsten Iteration?

## NASA-TLX Bewertung

> "Nun möchte ich Sie bitten, einen standardisierten Fragebogen namens *NASA Task Load Index* auszufüllen. Dies hilft mir, die wahrgenommene kognitive Arbeitsbelastung, die Sie bei der Nutzung des Systems erlebt haben, wissenschaftlich zu messen. Bitte bewerten Sie die sechs Dimensionen (Mentale Anforderung, Physische Anforderung, Zeitliche Anforderung, Leistung, Anstrengung und Frustration) basierend auf den Aufgaben, die Sie gerade abgeschlossen haben."

Vergleichen Sie es mit Ihnen bekannten Systemen und einer Vorstellung, wie die verschiedenen Modi Einfluss darauf haben könnten im Vergleich zu bestehenden Systemen.

*NASA-TLX auf Papier oder Tablet übergeben*

## Abschluss

- Fällt Ihnen noch etwas anderes ein? Gibt es etwas, worüber wir nicht gesprochen haben, das Sie erwähnen möchten?

*Aufnahme stoppen* • *Bedanken*

## A.3 Interview Guide Volunteer

**Bachelorarbeit:** "Volunteer Matching System"
**Datum:** 01.04.2026
**Institution:** Universität Zürich (UZH)

## Checkliste: Vor dem Interview

- Prototyp bereit?
- Raum für Studie bereit?
- Einverständniserklärung zweimal ausgedruckt und Stift verfügbar?
- Gerät für Aufnahme bereit / Teams-Aufnahme bereit?
- NASA-TLX (Papier oder online) bereit?
- Smartphone stummgeschaltet und auf Flugmodus?

## Einführung

- **Persönliche Vorstellung:** Nikola Petrovic, Major Softwaresysteme, Minor Neuroinformatik, schreibe meine Abschlussarbeit.
- **Einverständniserklärung:** Ich möchte das Interview für die spätere Analyse aufzeichnen. Alle Daten werden selbstverständlich vertraulich behandelt, anonymisiert und nur für wissenschaftliche Zwecke in diesem Projekt verwendet.
- Interview wird auf Hochdeutsch stattfinden, ist das ok?
- Ist es für Sie ok, wenn ich die Sitzung aufzeichne?

*Audioaufnahme einschalten / Auf Hochdeutsch wechseln*

- Geschätzte Zeit insgesamt: 30–50 min.
- **Sehr wichtig:** Keine falschen Antworten, alle Ihre Erkenntnisse sind wichtig für unsere Arbeit und werden uns helfen.
- Wenn Sie nicht antworten wollen, ist das natürlich auch ok.

## Einführung in das Projekt

- In vielen Krisen bieten viele Menschen sehr schnell Hilfe an, und Koordinatoren müssen Freiwilligen unter Zeitdruck Aufgaben zuweisen.
- Bestehende Tools sind dafür oft nicht ausgelegt und können viel mentale Arbeitsbelastung erzeugen — entweder ganz automatisch oder ganz manuell.
- In meinem Projekt entwickle ich einen Prototyp einer webbasierten Koordinationsplattform namens *Crisis Match*.
- Es konzentriert sich darauf, Koordinatoren bei der Verwaltung von Aufgaben und Freiwilligen zu helfen und die kognitive Belastung bei der Entscheidung zwischen verschiedenen Zuweisungsmodi (manuell, halbautomatisch, automatisch) zu reduzieren.

## Einführung in den Prototyp

- Es gibt im Wesentlichen zwei Ansichten, auf die Sie in der Anwendung zugreifen können:
  - Als **Freiwilliger** können Sie sich registrieren und für Aufgaben bewerben.
  - Als **Koordinator** können Sie Aufgaben erstellen und verwalten.
- In diesem Modus können Sie die verschiedenen Automatisierungsmodi nutzen.

*Nun möchte ich, dass Sie diese Dinge ausprobieren:*

- Sie übernehmen nun die Rolle eines Freiwilligen in einer fiktiven Krise.
- Ich werde mit Ihnen einen Durchlauf machen, indem Sie mehrere Aspekte des Prototyps durchgehen, wie man es in einer echten Situation machen würde:
  1. Sie werden sich einloggen und sich auf Aufgaben bewerben, die Sie interessieren.
  2. Ich werde Sie annehmen / ablehnen.
  3. Ich stelle Ihnen ein paar Fragen dazu.
- Ausserdem möchte ich Sie bitten, direkt laut auszusprechen, was Sie tun und was Sie denken.
- Also, zum Beispiel: “Ok, ich sehe diese Seite, ich denke, ich muss auf diesen Button klicken, damit ich auf diese Seite komme, aber das verwirrt mich ein bisschen, etc.”
- Wir gehen es gemeinsam durch.
- **Und wichtig:** Wir testen hier die App, nicht den Nutzer.

- Also, wenn etwas nicht klar ist, ist das ein Problem der App und ich möchte, dass Sie es mir sagen.
- Durchklicken des Prozesses der Erstellung des Kontos und Bewerben auf Aufgaben.

## Interview-Fragebogen

### Aktueller Bewerbungsablauf

- Welche Erfahrungen haben Sie bisher mit Freiwilligenarbeit (während einer Krise)?
- Was sind die Hauptherausforderungen bei der Bewerbung als Freiwilliger in verschiedenen Organisationen?
- Wie, wenn überhaupt, beeinflusst Zeitdruck diesen Prozess?
- Wo liegen Ihrer Erfahrung nach die grössten Engpässe oder Herausforderungen in diesem aktuellen Aufbau?
- Was hat Sie damals, falls es etwas gab, besonders gestört?

### Wahrgenommene Benutzerfreundlichkeit & UI-Design

- Wie haben Sie den Arbeitsablauf über die Bewerbung hinweg erlebt?
- Wie empfanden Sie die Datenmenge, die man angeben musste?
- Wie haben Sie sich beim Ausfüllen der Angaben gefühlt?
- Gab es Momente, in denen Sie unsicher waren, was als Nächstes zu tun ist? Sagen Sie mir, wo das passiert ist.
- Was würden Sie ändern / anpassen (UI-Design)?

### Kognitive Belastung & Entscheidungsermüdung

- Was denken Sie, ist essenziell für so eine Plattform, aus der Freiwilligenansicht?
- Gab es Momente während der Simulation, die sich überwältigend oder anspruchsvoll anfühlten? Erzählen Sie mir davon.

### Vertrauen & Human-in-the-Loop (Strukturelles / Psychologisches Empowerment)

- Erzählen Sie mir von einer Erfahrung mit Freiwilligenarbeit, die Ihnen in Erinnerung geblieben ist, bezogen auf die Administration.

### Allgemeines Feedback

- Wenn dieses System morgen in einer echten Krise eingesetzt würde, was ist das grösste Risiko oder die fehlende Funktion, die Sie sehen? Und was ist Ihre oberste Priorität zur Verbesserung dieses Prototyps in der nächsten Iteration?

## NASA-TLX Bewertung

> "Nun möchte ich Sie bitten, einen standardisierten Fragebogen namens *NASA Task Load Index* auszufüllen. Dies hilft mir, die wahrgenommene kognitive Arbeitsbelastung, die Sie bei der Nutzung des Systems erlebt haben, wissenschaftlich zu messen. Bitte bewerten Sie die sechs Dimensionen (Mentale Anforderung, Physische Anforderung, Zeitliche Anforderung, Leistung, Anstrengung und Frustration) basierend auf den Aufgaben, die Sie gerade abgeschlossen haben."

Vergleichen Sie es mit Ihnen bekannten Systemen und einer Vorstellung, wie die verschiedenen Modi Einfluss darauf haben könnten im Vergleich zu bestehenden Systemen.

*NASA-TLX auf Papier oder Tablet übergeben*

## Abschluss

- Fällt Ihnen noch etwas anderes ein? Gibt es etwas, worüber wir nicht gesprochen haben, das Sie erwähnen möchten?

*Aufnahme stoppen* • *Bedanken*